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RSM Optimized Feasibility Assessment of Hybrid Algae-Plastic Oil Blended Diesel Fuel with Binary Additives CuO−MgO on Performance and Emissions in CRDI Diesel Engine

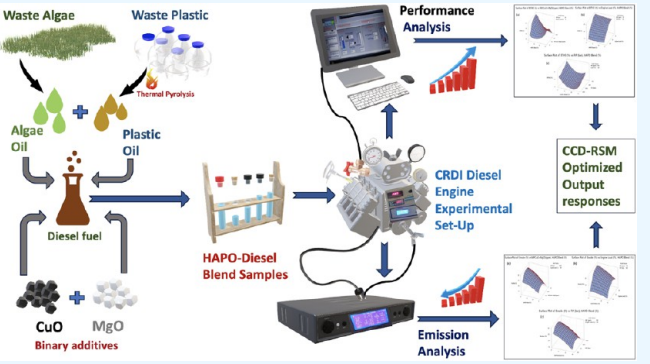
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ABSTRACT: This study was carried out to analyze the feasibility of a less-explored alternative fuel blend combination for diesel engines comprising third-generation biofuel (algae oil (AO)) and fuel derived from waste plastic. Combining these two oils in equal proportions, hybrid algae-plastic oil (HAPO) blends were formed. For this study, waste plastic oil (WPO) was produced through a lab-scale thermal pyrolysis process. The impact of different blend ratios (0−40 vol %) of HAPO-blended diesel fuel along with input engine load (20−100%) and fuel injection pressure (FIP) (300−700 bar) is experimentally tested for its performance and emission characteristics in a single-cylinder common-rail direct-injection (CRDI) diesel engine. The impact of combined total binary (CuO + MgO) metal oxide nanoadditives (0−300 ppm) in a 1:1 fixed mass ratio formulation with HAPO blends, along with engine key input variables, has also been assessed. The response surface methodology (RSM) technique was applied for optimization by developing trials using a four-factor central composite design (CCD) approach, and output responses were analyzed using surface plots. The R^2 values of estimated outputs for brake thermal efficiency (BTHE), exhaust gas temperature (EGT), nitrogen oxide (NO_x), smoke, and unburned hydrocarbon (HC) were determined to be nearly approaching 90−99.9%, except for the limited predictive R^2 for the carbon monoxide (CO) model. The outcomes from the input parameters were optimized for maximum performance and minimum emissions. RSM-predicted optimized input value settings at 40 vol % HAPO blend, 300 ppm of total binary additive with 441.4 bar of FIP, and 52.32% engine load have given a significant RSM-predicted response of 20.27% of BTHE and 459.36 K of EGT with noticeable reduction in smoke by 11.5%, HC by 16.1 ppmv, NO_x by 358.33 ppmv, and CO by 0.0927 vol % with an overall composite desirability of 81.98% within the investigated experimental domain. Final tests were conducted for experimental verification of optimized conditions by fixing the input parameters to optimized values to confirm authenticity. The percentage error between RSM-predicted values and experimental values for BTHE, EGT, smoke, and CO output responses was less than 5%, whereas for HC and NO_x, the errors were 5.93 and 5.21%, respectively. The results demonstrate the short-duration laboratory-scale technical feasibility and promising performance of characterized HAPO-blended diesel as a candidate fuel for CRDI diesel engines with significant tailpipe emission reduction potential.



1. INTRODUCTION

The world's economy is growing rapidly, which increases energy demand with every passing decade due to industrialization, technological development, and population growth. Most transportation, industry, agriculture, and, in some regions, even commercial electricity production around the world rely on conventional fuels derived from petroleum. However, their heavy use significantly contributes to air pollution and greenhouse gas accumulation.[1] The information from the U.S. Energy Administration projects that worldwide liquid fossil fuel consumption will rapidly increase from around 86 million barrels per day in 2008 to more than 112 million barrels per day by 2035.[2] This heavy reliance on fossil resources has produced significant environmental problems and has accelerated efforts toward renewable and sustainable energy alternatives. Biodiesel is a practically and ecologically viable alternative to petroleum diesel that combines reduced emissions of pollutants with high biodegradability.[3] Biodiesel, based on transesterification of oils or fats rich in triglycerides

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Table 1. Review on Recent Optimization Studies on Engine Testing and Biofuel Production with Nanoadditives

authors and year	optimization method	blend ratio and fuel type	nanoadditives with concentration	impact on engine performance or fuel production	impact on emissions or fuel quality	diesel engine type	result validation
Savaş et al.[12]	RSM	diesel with jojoba biodiesel (10−30%)	$MgCO_3$ (50−150 ppm)	minimum BSFC achieved with 75 ppm approx	CO and HC emissions reduced	one cylinder, DI fuel, 3.2 kW @ 3000 rpm	error rates <10%, R^2 > 95%
Yeneneh and Sufe[13]	ANN-based model	waste plastic biodiesel and diesel (50%)	Al_2O_3 (50 ppm)	BTHE increases by 7.2% by adding Al_2O_3	NO_x increased by 6.08%, HC reduced by 37.84%, CO reduced by 50%	single-cylinder diesel engine	R^2 > 0.999
Yeneneh and Sufe[14]	RSM optimized	cottonseed oil biodiesel	nickel-doped eggshell CaO	FAME yield of 98.01%	low moisture content of 1.2% and high volatility	no CRDI study, biodiesel production	R^2 = 0.9953
Muniyappan and Krishnaiah[15]	RSM	mahua biodiesel (10−30%)	CuO (copper oxide) (25−75 ppm)	BTHE increases values to 34.15%, 4.6% higher, approx. 0.23 kg/kWh lower BSFC	reduction of 22.2% in HC, 25.6% in CO, and 19.9% in smoke NO_x increases	VCR, 1-cylinder, 4-S, Technomech eddy dynamo @ 1500 rpm	MAPE of 3.0962%
Singh et al.[16]	CCD-RSM	ternary blends of biobutanol, biodiesel, and diesel (10, 20, and 70%)	ZnO (zinc oxide) 0−100 ppm	higher ignition delay increases combustion and performance, BTHE increases, BSFC decreases	lower CO, HC, and smoke	air-cooled VCR engine (16−20 CR)	error < 5%
Sunil Kumar et al.[17]	RSM, ANOVA with deep learning	waste plastic oil (7−13%), 1-butanol (3−7%) with diesel (80−90%)	TiO_2-dopped butanol (25−100 ppm)	BTHE increases values to 37.3%, IMEP dominated	8% CO_2 and 1800 ppm of NO_x reduction, HC and CO also reduce	RTD-type, water-cooled, single-cylinder diesel engine @ 1500 rpm	high R^2, lower MSE
Birhanu et al.[18]	RSM optimized	Ethiopian *Podocarpus falcatus* seed oil biodiesel	CaO-CeO_2	fuel production yield was 98.32%	acceptable fuel oxidative stability, no emission testing	production application, no engine study	R^2 = 0.9916
Kumar et al.[10]	muti-objective RSM (CCRD matrix)	PPO (plastic pyrolytic oil) (0−40%)	Al_2O_3 (aluminum oxide) (0−100 ppm)	BTHE and BSFC both increase until optimal values only	CO reduces to 0.1%, HC to 35.4 ppm, and NO_x by 562 ppm	Kirloskar 1-C, 4-S, VCR, water-cooled DI diesel engine, 1500 rpm, 3.7 kW	D = 0.89, error < 5%
Chetia et al.[19]	BBD-RSM	waste cooking biodiesel	CeO_2 (cerium oxide) (50−100 ppm)	lower performance for BTHE and BSFC but better combustion	11.5% reduction in CO, 7.9% in HC, and 20% in smoke reduction	1-C, DI diesel engine @ 1500 rpm, 3.5 kW	error = 7%
Uslu[20]	RSM	waste cable pyrolysis oil (20% WCPO and 80% diesel)	B_2O_3 (boron oxide) (20−60 ppm)	reduced BTHE by 8% and BSFC reduced by 3.8%	HC lowered by 27.2%, CO by 6%	Lutian LT178F @ 3000 rpm, air-cooled, 1-C	D = 0.72, error = 4.8%
Uslu et al.[21]	multiresponse RSM	spirulina microalgae oil with diesel (20% SSMB/80% diesel)	Fe_3O_4 (iron-oxide) (25−75 ppm)	BTHE improved by 13%, BSFC deteriorated by 10%	CO and HC reduced by 20 and 23%; CO_2 and NO_x increased by 12 and 22%	Kirloskar water-cooled, VCR engine, 1-C @ 1500 rpm, 3.7 kW	D = 0.83, error ≤ 6.5%
Rajavel et al.[22]	hybrid optimization RSM-ANN-GA	waste cooking oil (20%) with diesel (B20, B20GS50)	green ZnO (zinc oxide from aloe vera)	BTHE improves and values to 31.3%	NO_x reduction by EGR, values to 277 ppm	DI-CI engine, 1-C, 4-S, Kirloskar AV1 type	R^2 = 99%
Canan[23]	multipurpose RSM	waste cable pyrolysis oil (20% WCPO, 80% diesel)	graphene (20−40 ppm)	BTHE increases, optimally values to 23%; BSFC reduces, values to 344 g/kWh	CO and HC reduced, CO_2 and NO_x increased	Lutian 3 GF-ME, air-cooled engine @ 3000 rpm	R^2 = 96.5−99.8%
Savaş et al.[24]	RSM	jojoba biodiesel (10%) with diesel	MgO (50−150 ppm)	BSFC minimum @ 41 ppm of MgO at 1.5 kW, 6.7% reduction in BSFC, 7.3% increase in BTHE	CO and HC decreased by 13 and 30%, NO_x and CO_2 increased	1-C, DI, 4-S, CI engine, 3.2 kW @ 3000 rpm	error < 6.65% R^2 = 95.65, D = 0.75
Şener et al.[25]	RSM-MOO	J10 (jojoba biodiesel 10% with 90% diesel)	maghemite (Fe_2O_3) (50−150 ppm)	BSFC reduced by 4%; BTHE increased by 4.2%	CO and HC decreased by 49.8 and 66.9%; NO_x and CO_2 increased by 24 and 26%	Lutian 3 GF-ME, air-cooled engine @ 3000 rpm, 3.2 kW	error < 8%
Avcı and Yavaşoğlu[26]	RSM/CCD	Ni_2O_3 (nickel III oxide) (0−100 ppm)	biodiesel-*n*-butanol > (B20, B20But10)	increase in BTHE and BSFC reduces	13−28% reduction in HC, 8−43% in smoke, and 12−21% in NO_x	one cyl., 4-S, water-cooled, DI diesel engine	R^2 = 91−99%
Sujin et al.[27]	BBD, MLT DBN, and hybrid DBN	CaO-TiO_2 (5−20 ppm)	mixed nonedible oil biodiesel (10−40%)	BTHE and BSFC both improved	NO_x and CO decrease	4-S, 1-C, VCR engine	R^2 = 99.8%

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with short-chain alcohols, is similar to diesel fuel in many ways but provides environmental advantages, such as reduced carbon monoxide (CO), hydrocarbon (HC), and particulate matter emissions.[4] Due to its oxygenated nature, nontoxicity, and lack of sulfur and aromatics, biodiesel has the potential to enhance combustion quality and reduce the overall carbon footprint of engines. It also contributes to rural development and energy independence through decentralized fuel production.[5]

However, large-scale biodiesel applications of edible oils, such as soybean, sunflower, and palm oils, raise serious concerns regarding food security and competition for agricultural land, commonly known as the food-versus-fuel issue.[6] To overcome these challenges, research has now focused on nonedible and waste-oil-based feedstocks as promising alternatives. In this regard, algae oil (AO) has gained much attention because of its efficiency and renewable nature. Algae may be grown on nonarable land using saline or wastewater, while specific species exhibit lipid contents over 40% of their dry mass.[7] The use of waste algal biomass, recovered from either industrial effluents or wastewater treatment processes, further reduces feedstock cost and supports the circular economy by converting biological wastes into energy resources.[8] Algae oil biodiesel has favorable properties, such as a high cetane number, oxygen enrichment, and good combustion efficiency. However, due to its relatively higher viscosity and moderate calorific value, pure algae oil application may slightly affect engine performance.[9]

Another major problem that the world is currently facing is the enormous heaped mountains of plastic waste. In today's production market scenario, plastic is a boon to society, but its nonbiodegradability has resulted in billions of tons of unused plastic waste. Just like the pollution from fossil fuels, plastic pollution has become a matter of concern for many years throughout the world.[10] As plastic is also a petroleum product with long hydrocarbon chains, it can be broken down through thermal cracking by a process called pyrolysis. Through pyrolysis, waste plastics can be converted into liquid fuel, commonly known as plastic pyrolytic oil (PPO) or waste plastic oil (WPO). Using this plastic oil (PO) derived from waste plastics through pyrolysis contributes as a solution to two major concerns of countries worldwide: first, providing an economical alternative to conventional fuels, and second, reducing the huge piles of plastic waste into a usable energy resource.[10,11]

However, plastic oil use in engines has certain limitations, restricting its widespread usage. Blending algae oil with plastic oil, a pyrolysis-derived hydrocarbon liquid prepared from waste plastics, is a synergistic strategy. The blend is termed as hybrid algae-plastic oil (HAPO) blend in this article. Plastic oil can enhance volatility and energy density, whereas algae oil can improve combustion quality owing to its oxygenated structure.[28] Therefore, the combination creates a balanced fuel blend using two major streams of wastes, solving the problems of both algae residue management and plastic pollution, one being nonbiological waste and the other biological. Algae oil is considered a biofuel, but plastic oil is a waste-to-energy resource, making this blend a hybrid.

Table 1 shows the review on the impact of various fuel blends and nanoadditives on performance and emissions from diesel engines using optimization techniques. Although various researchers have assessed biodiesel blend combinations from nonedible sources, including Jatropha,[29,30] Karanja, Mahua, and Eucalyptus oils, studies related to algae oil and plastic oil blended together are rarely available. Moreover, detailed optimization work studying the combined effects of algae oil and plastic oil blends on diesel engine performance is scarce. Most previous studies changed one parameter at a time, which limits knowledge on the complicated interactions of operating variables such as blend percentage, fuel injection pressure (FIP), exhaust gas temperature (EGT), and engine load. Sophisticated statistical tools like response surface methodology (RSM) provide a structured way of investigating and optimizing these multivariate relationships efficiently.[31] However, optimization analysis and applications of the combined algae and plastic fuel with diesel are still scarce. For further reduction of emissions from the engine, adding nanoparticles (NPs) with the fuel blend has been a promising practice, giving improved results in previous studies.[32] Metal oxides as nanoadditives have shown significant results in improving emission characteristics,[33] but combining these two commonly available metal oxides, CuO and MgO, is not fully explored. Therefore, these two have been added as novel nanoparticle catalyst combinations with HAPO blends.[31,32]

Previous studies have separately investigated these waste plastic-derived fuels, algae biodiesel, metal oxide additives, and RSM optimization for biodiesel production or engine optimization studies. Recent studies have reported nanoparticle-based waste plastic oil fuel blends with experimental and predictive modeling. In a recent investigation, Yeneneh and Sufe evaluated waste plastic oil and biodiesel blends with Al_2O_3 as a nanoadditive in a single-cylinder diesel engine. They analyzed performance, emission, and combustion characteristics with artificial neural network modeling depicting the predictive approach for engine optimization.[13] However, the present study implies a different modeling approach, nanoparticle combination, input engine variables, and output response.

RSM modeling is also being used to optimize production processes of biodiesel. Birhanu et al. have applied RSM optimization in the transesterification of *P. falcatus* seed oil with a binary CaO-CeO_2 catalyst and evaluated oxidative stability and fuel quality.[18] Similarly, Yeneneh and Sufe used an RSM model for cottonseed oil biodiesel production along with a Ni-doped eggshell CaO catalyst. However, the focus of these studies was on the use of RSM for biodiesel production yield and catalyst performance.[14]

In contrast, the present study investigates and optimizes the performance and emission characteristics of a common-rail direct-injection (CRDI) diesel engine running on a rarely explored oil combination of hybrid algae-plastic oil (HAPO) blended diesel fuel with the binary metal oxide nanoadditives CuO and MgO in a fixed 1:1 formulation. Characterization of both novel combinations of fuel blends and nanoadditives is done by Fourier transform infrared (FTIR), transmission electron microscopy (TEM), and physicochemical analysis. It analyzes the blend ratio, intensity of nanoparticles (NPs) added, fuel injection pressure (FIP), and engine load using a central composite design (CCD) approach under the RSM framework. Key response parameters in the evaluation are brake thermal efficiency (BTHE), EGT, smoke, CO, unburned HC, and nitrogen oxide (NO_x) emissions. Thus, this work's novelty and contribution lie in the unexplored integrated evaluation of a hybrid fuel blend of waste plastic and algae oil with binary additives of CuO and MgO. The results aim to find the optimal settings in engine operating conditions that give

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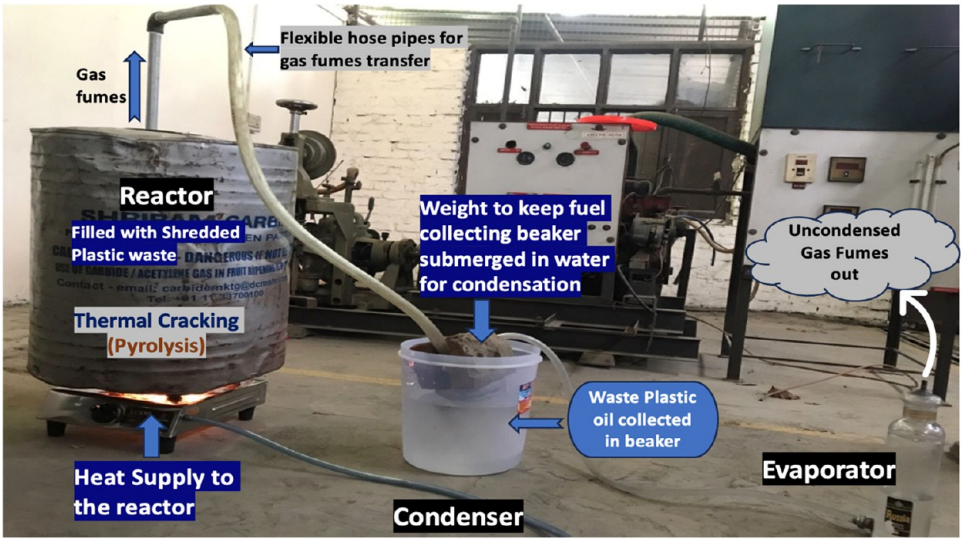


Figure 1. Pyrolysis process of plastic oil preparation from the mixed waste plastic feedstock.

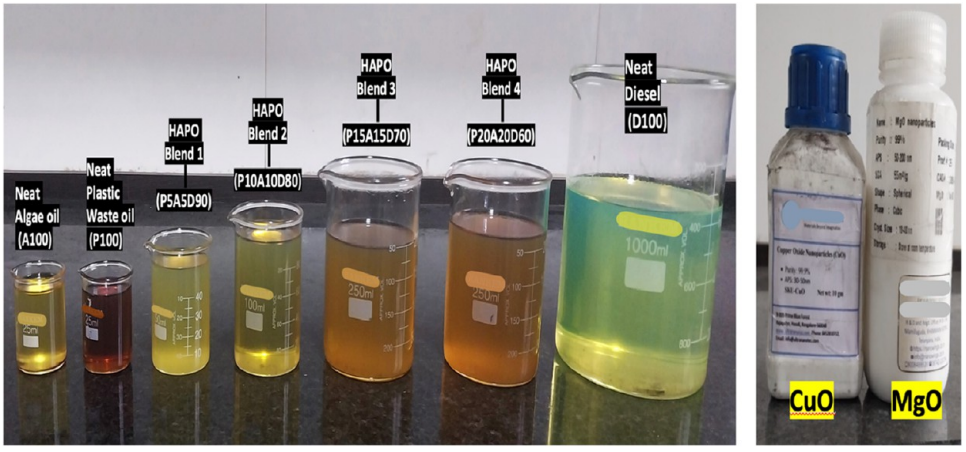


Figure 2. Materials used for fuel blend preparations: (a) prepared HAPO blends with varying vol %. (b) CuO and MgO nanoparticles.

higher efficiency with lower tailpipe emissions and hence establish HAPO-blended diesel as a laboratory-scale technically feasible alternative to conventional diesel fuel.

2. MATERIALS, CHARACTERIZATION, AND EXPERIMENTAL SETUP

The fuel blend used in this study is a novel hybrid algae-plastic oil (HAPO) blended with commercial diesel. Plastic oil was extracted from the waste plastic within the university engine research lab, and the algae oil was purchased from an authentic vendor with its chemical certification of purity. Both of the oil samples were tested and characterized by standard certified laboratories. For diesel engine compatibility of these HAPO blends, a VCR engine experimental setup is used to carry out performance and emission analysis by adding the metal oxide nanoparticles (CuO and MgO) sourced from the chemical supplier in the nearby market. All of the appropriate checks and validations are done to confirm the genuineness of the materials used in the study.

2.1. Pyrolytic Plastic Oil Extraction

For plastic oil extraction, the mixed waste plastic bottles were obtained from the university waste collection center in Delhi, which is a mixed-grade plastic made up of HDPE mainly with some LDPE and PS fractions, and were washed, dried, and shredded into small pieces of approximately 10−20 mm size and filled in a cylindrical alloy steel container (reactor). Laboratory-scale thermal pyrolysis was performed under limited oxygen conditions using a gas stove as a heating source.

The thermal cracking or breakdown of the polyethene compounds at high temperature with almost the absence of oxygen is thermal pyrolysis.[11,33] No catalyst was used in the production of the product. The evolved gas fumes were transferred through flexible hose pipes to a transparent beaker kept in another container filled with water for gas fume condensation. The condensed oil gets collected in the beaker, and the remaining uncondensed gas fumes get exhausted from the hose pipe connected to the evaporator. The yield for this lab-scale thermal pyrolysis process was around 68% by the mass of the initial plastic feedstock. Produced plastic oil cannot be considered as a uniquely reproducible pyrolysis product. Plastic oil was collected only for the sample HAPO blend preparation. Reactor temperature, heating rate, and vapor residence time were not recorded during the pyrolysis experimental procedure. Therefore, the produced oil was characterized experimentally before fuel blend preparation and engine testing. This process with the entire setup was carried out in the engine research lab of the university itself, as shown in Figure 1, with a reactor, condenser, and evaporator.

2.2. HAPO Fuel Blend Formation with Metal Oxide Nanoparticles

HAPO-blended diesel fuel was prepared by mixing plastic oil (PO) derived from waste plastics through a thermal cracking or thermal pyrolysis process with algae oil (AO) in equal proportions for every blend. These hybrid blends combine the benefits of the energy density

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of PO[37] with the benefits of the combustion quality of AO due to its oxygenated structure.[6,38] Both oils were stirred multiple times at blending stages for homogeneous mixing with the magnetic stirrer. One stirring was done before blending with diesel fuel for about 25 min at 1500 rpm for each blend ratio and experimental trial, and another was done after the addition of metal oxide additives (NP), i.e., combined binary (CuO + MgO in equal 1:1 mass ratio) along with diesel fuel in the desired ratios as per the design of the experiments. For example, a total of 300 ppm of binary additive corresponds to 150 ppm of CuO and 150 ppm of MgO. Equal proportions of both metal oxide nanoparticles and both fuel types were selected as an initial balanced formulation to maintain a fixed concentration of additives and fuels in blends with different physicochemical properties. The equal ratio of 1:1 was kept to make the experimental matrix feasible in a four-factor CCD-RSM model and to avoid complexity by not increasing the number of variables related to fuel blend and additive concentration. Focus was kept on those input variables that were considered to be essential for assessing engine performance and emissions using these fuel-additive blend combinations. However, variation in proportions of fuel and additives in the blend could be another possible analysis.

To achieve the homogeneity and initial dispersion of these nanoparticles into the HAPO-blended diesel at total binary additive concentrations of 75, 150, 225, and 300 ppm during the short-duration laboratory experiment, magnetic stirring at the second stage was done for 1.5 h immediately before the engine testing.[39] Ultrasonication and chemical dispersants were not used in the present study, so long-term colloidal stability was not assessed. Visual inspection of these nanoparticles showed negligible sedimentation within the fuel blend during the testing interval. Figure 2b shows CuO and MgO procured for the use as nanoadditives in blends, and Table 2 presents specifications and properties of these nanoparticles.

Table 2. Designation of HAPO Test Fuel Blends

HAPO fuel blend	designation
P5A5D90	plastic oil 5% + algae oil 5% + diesel 90%
P10A10D80	plastic oil 10% + algae oil 10% + diesel 80%
P15A15D70	plastic oil 15% + algae oil 15% + diesel 70%
P20A20D60	plastic oil 20% + algae oil 20% + diesel 60%

HAPO blends were prepared in a ratio of vol % of the total blended fuel. The vol % taken in the trials are 10, 20, 30, and 40% along with the neat diesel. Figure 2a shows the samples prepared by the neat algae oil (A100) and neat plastic oil (P100) taken in equal portions and blended with the neat diesel in the above-mentioned vol %.

Thereby, four different blend samples, named P5A5D90 for 10% HAPO blend volume, P10A10D80 for 20% HAPO blend volume, P15A15D70 for 30% HAPO blend volume, and P20A20D60 for 40% HAPO blend volume, were prepared, where P, A, and D refer to plastic, algae oil, and diesel subsequently followed by its vol % with the neat diesel (D100). Table 2 presents designation of the test fuel HAPO blends formed.

2.3. Fuel and Nanoparticles Characterization

The physicochemical characterization of laboratory-produced plastic oil, commercially sourced algae oil, and prepared HAPO blends, along with metal oxide nanoadditives used, is carried out as per the ASTM standards for understanding the physical and chemical attributes of these materials. Use of metal oxide nanoparticles is associated with enhanced combustion and lesser ignition delay. The commercially supplied and reported particle size information was used for initial powder selection. Table 3 shows the specifications and properties of CuO (copper oxide) and MgO (magnesium Oxide) sourced from Sigma-Aldrich with almost 99.9% purity and the supplier stated an average particle size of nanoscale. For identification of morphology related to the shape and size of these nanoparticles, TEM (transmission electron microscopy) is utilized in this study. TEM analysis revealed nonuniform particle size from nanoscale to sub-

Table 3. Specification of Copper and Magnesium Oxide Nanoparticles

property	CuO	MgO
color	black	white
density (g cm^{-3})	6.25	3.57
diameter	>70 nm	>50 nm
structure	monoclinic	cubic
melting point	1326 °C	2852 °C

micrometer for MgO ranging from 89.9 to 130 nm and from 133 to 240 nm for CuO. FTIR (Fourier transform infrared) spectroscopy was done for all base fuel oils and both nanoadditives. Table 4 shows

Table 4. Comparison of Physicochemical Properties among HAPO Blends with Diesel[a]

properties	density at 25 °C	higher heating value	flash point	cetane number	kinematic viscosity @ 40 °C
unit	kg m^{-3}	kJ kg^{-1}	°C		mm^2 s^{-1}
ASTM standard	D287	D4809	D93-58T	D976	D445
D100	820	44,661	56	58	2.09
algae oil	874	40,517	161	168	5.43
plastic oil	780	43,486	35	38	2.27
P5A5D90	821	44,112	54	56	2.38
P10A10D80	823	43,761	51	55	2.44
P15A15D70	826	43,308	49	52	2.57
P20A20D60	819	42,976	53	50	2.79

[a]Higher heating values correspond to the base fuels and HAPO−diesel blends before the nanoparticle's addition.

the comparison of all major physical and chemical attributes as per the ASTM standards for both the base oils and their four HAPO blends with diesel oil before the nanoparticle's addition. The fuel property measurements were carried out with specified calibrated laboratory instruments; the higher heating value of each fuel sample was measured using a calibrated bomb calorimeter.

2.3.1. FTIR Spectroscopy of Algae and Plastic Oil. Fourier transform infrared spectroscopy was carried out for laboratory-produced plastic oil, commercially sourced algae oil, and additives (CuO + MgO) used in this experimental study.

The FTIR spectral analysis of the pyrolysis oil obtained from waste plastic was carried out across a broad wavenumber range of 4000−400 cm^{-1} to identify the key functional groups present in the sample, as shown in Figure 3. Three closely spaced absorption peaks appearing at 2953.03, 2921.31, and 2852.34 cm^{-1} drew immediate attention, as these are well-known signatures of aliphatic C−H stretching, with the first peak belonging to methyl groups and the latter two peaks are of methylene chains.[40] Such a pattern strongly points toward a fuel matrix built around long, saturated hydrocarbon backbones, not unlike what one would expect from a diesel or kerosene fraction.[41] Moving into the mid-infrared fingerprint region, two moderately intense bands were recorded at 1456.35 and 1376.29 cm^{-1}, which reflect the bending and deformation of $-CH_2-$ and $-CH_3$ units, respectively.[42] These features, taken together with the stretching bands in the 3000 cm^{-1} region, paint a consistent picture of a paraffinic-rich oil.[40] A relatively weaker band near 944.74 cm^{-1} hints at some degree of unsaturation, likely from terminal olefinic groups that survived the pyrolysis process.[43] The absorption features clustered around 735.51 and 697.41 cm^{-1} are suggestive of methylene rocking modes and possibly some residual aromatic or vinyl-type C−H out-of-plane deformations, though their low intensity implies these are minor constituents rather than dominant species.[44] Toward the far end of the spectrum, faint bands near 473, 458, and 418 cm^{-1} were observed, most likely arising from skeletal bending motions of the carbon framework. the most explanatory observation was what the

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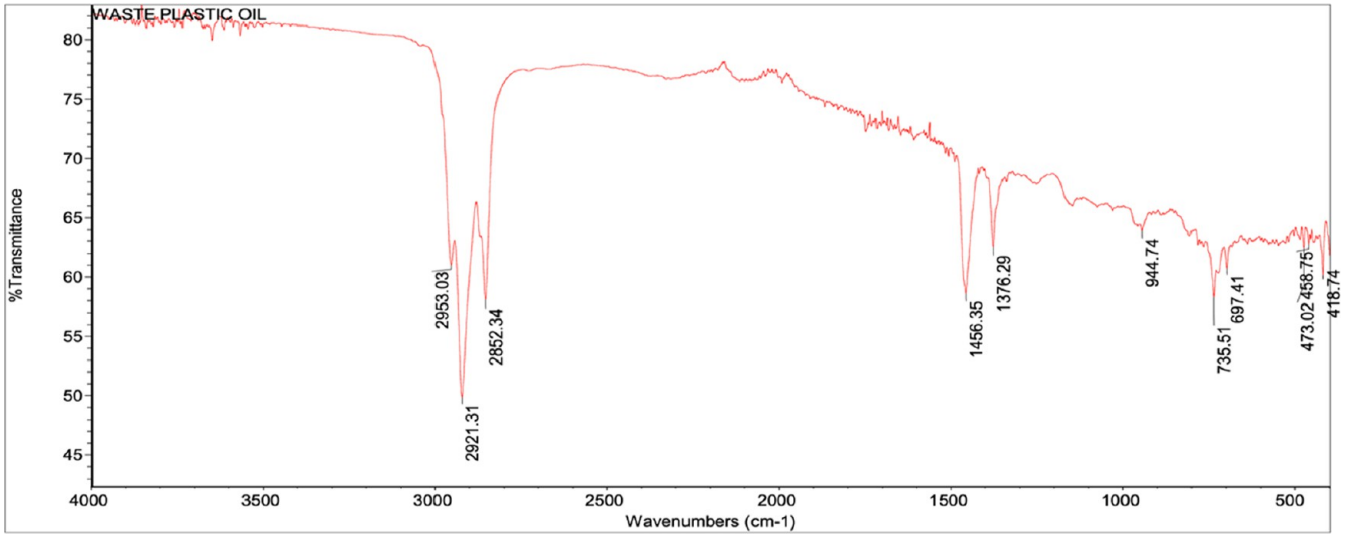


Figure 3. FTIR spectrum of waste plastic pyrolysis oil over 4000−400 cm^{-1} showing aliphatic C−H band stretching.

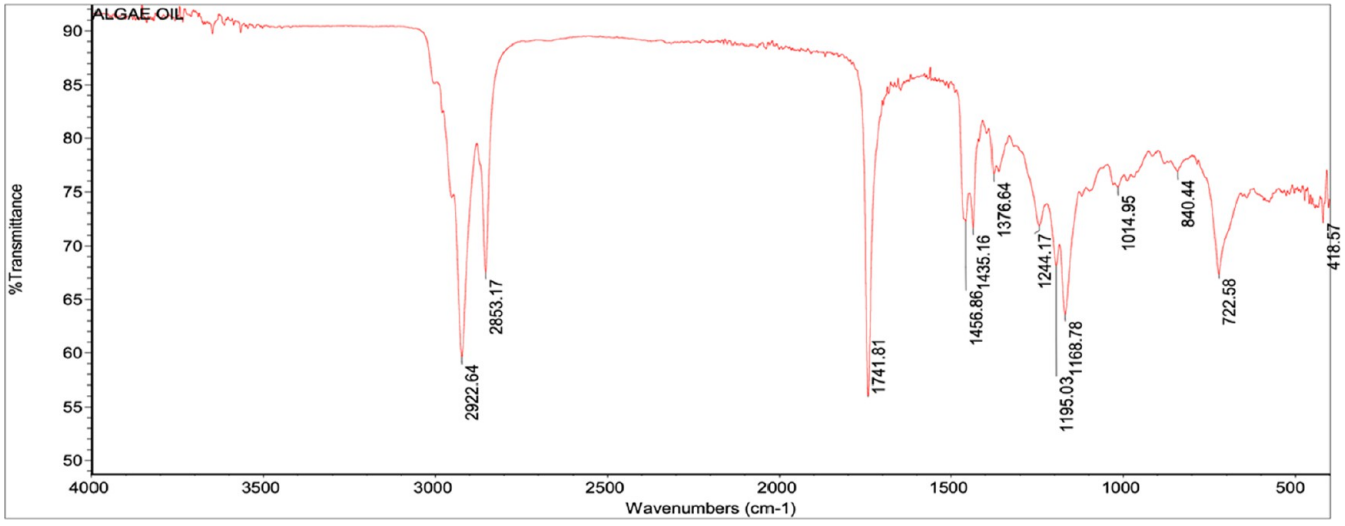


Figure 4. FTIR spectrum of algae oil showing a carbonyl absorption band near 1741 cm^{-1}.

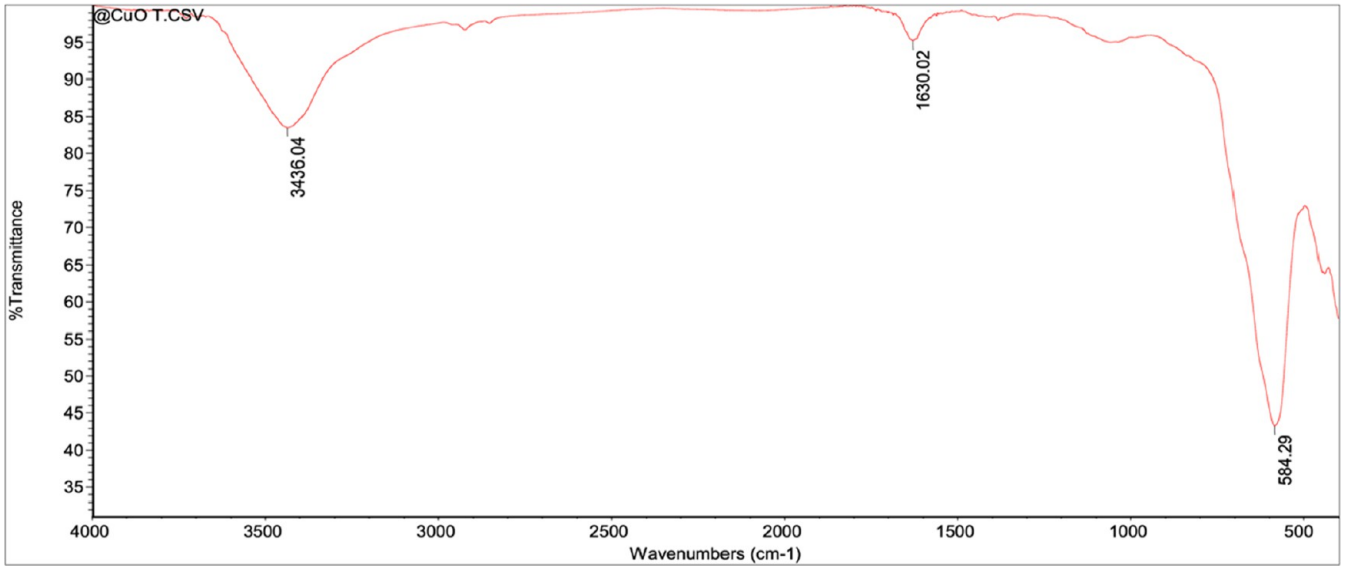


Figure 5. FTIR spectrum of CuO nanoparticles, including the metal−oxygen vibration region.

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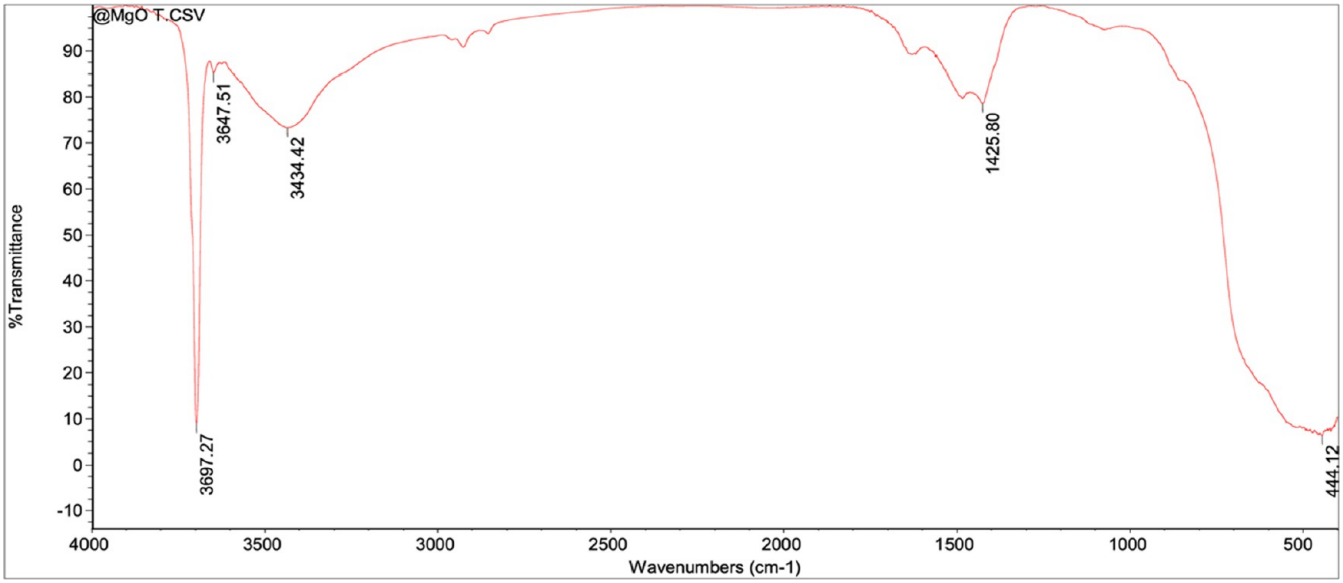


Figure 6. FTIR spectrum of MgO nanoparticles, including low-wavenumber Mg−O vibration region.

spectrum did not show, there was no broad hydroxyl band in the 3200−3500 cm^{-1} range, and no discernible carbonyl peak around 1720 cm^{-1}, which collectively rule out any significant water contamination or oxygenated impurities in the oil.[42,43] This absence is particularly meaningful from a fuel quality standpoint, as it may indicate low corrosive potential and better thermal stability. Taken as a whole, the FTIR data affirm that the waste plastic-derived oil is predominantly hydrocarbon in nature, bearing a chemical character that aligns well with conventional fossil fuels and therefore holds genuine promise as a viable alternative fuel source.[40,41,43−45]

When the FTIR spectrum of algae-derived oil was examined and shown in Figure 4 alongside that of the waste plastic oil, several interesting similarities and contrasts came to light. Much like the plastic-derived sample, the algae oil spectrum recorded over the same 4000−400 cm^{-1} range displayed prominent C−H stretching absorptions in the 3000 cm^{-1} region, with peaks appearing at 2922.64 and 2853.17 cm^{-1}, corresponding once again to the asymmetric and symmetric stretching of methylene groups within aliphatic hydrocarbon chains.[46] This overlap with the plastic oil spectrum is not incidental; it reflects the shared hydrocarbon backbone that makes both oils thermodynamically interesting as fuel candidates. However, the resemblance begins to diverge quite noticeably as one moves further into the mid-infrared region. The most striking difference is the appearance of a sharp, well-defined absorption band at 1741.81 cm^{-1} in the algae oil spectrum, a feature entirely absent in the waste plastic oil. This band is a textbook indicator of ester carbonyl (C═O) stretching, and its presence here strongly suggests that the algae oil retains a meaningful proportion of triglycerides or fatty acid ester linkages, a chemical signature that is entirely consistent with the lipid-rich biological origin of microalgae.[47]

Both waste plastic oil and algae oil have the C−H stretching bands appearing close to 2922 and 2853 cm^{-1}, common to both samples, confirming that aliphatic hydrocarbon chains form the structural backbone of each oil.[43,48] The waste plastic oil showed no trace of oxygenated groups, no hydroxyl, no carbonyl, which essentially means it is a clean hydrocarbon mixture, compositionally close to conventional diesel.[41] The algae oil, on the other hand, carried a distinct carbonyl absorption at 1741 cm^{-1}, accompanied by C−O stretching features around 1244, 1195, and 1168 cm^{-1}, collectively pointing to the presence of fatty acid esters and triglycerides rooted in its biological origin.[49,50] In practical terms, waste plastic oil appears more directly usable as a drop-in fuel, while algae oil, despite its rich lipid content and oxygenated groups, would benefit as a pathway toward promising hybrid fuel blend.

2.3.2. FTIR Spectroscopy of CuO and MgO Nanoparticles. The characterization of metal oxide nanoparticles, CuO and MgO, was done through FTIR spectroscopy, as shown in Figures 5 and 6.

The peaks in the FTIR spectrum of CuO nanoparticles in Figure 5 show larger band stretching of the oxygen−hydrogen bond (O−H) of around 3436.04 cm^{-1}, but for the H−O−H bond, the peak is near 1630.02 cm^{-1}, which represents the presence of the hydroxyl group along with surface-adsorbed water,[51] followed by another specific peak intensity of around 584.29 cm^{-1} corresponding to lattice vibrations of copper oxide. The entire spectrum confirms the formation of monoclinic copper oxide. This supports and verifies that the purity and chemical identity of these CuO nanoparticles may be suitable as additives, indicating better combustion characteristics.[52]

In the MgO FTIR spectrum shown in Figure 6, the recorded range was 4000−400 cm^{-1}, which is centered at 3697.24 cm^{-1}, depicting a deep but narrow valley pattern for O−H bond stretching vibrations along with a lesser broadband near 1425.80 cm^{-1} for H−O−H bond bending of physically adsorbed water molecules.[53] A significant absorption band is observed near 444.12 cm^{-1}, with features within the range of 500−800 cm^{-1} assigned to magnesium oxide bond stretch with lattice vibrations that confirm the hexagonal or cubical structure.[34]

Both the peaks of MgO and CuO obtained through FTIR spectroscopy reveal that these metal oxide nanoparticles enhance combustion characteristics by promoting complete fuel oxidation and better dispersion, increasing overall performance and reducing emissions.

2.3.3. Characterization of Copper and Magnesium Oxide Nanoparticles. Transmission electron microscopy was carried out to get a closer look at the physical characteristics of the synthesized CuO and MgO nanoparticles before their use as fuel additives. The micrograph was captured at 30,000× magnification and 120 kV accelerating voltage, as shown in Figure 7.

The TEM micrograph image of CuO shown in Figure 7a,b revealed that the particles largely adopt a cubic or near-rectangular shape, with some mildly irregular edges along their periphery, a morphological trait fairly typical of CuO nanostructures produced via chemical synthesis methods.[54] Direct measurements from the image placed individual particle sizes between roughly 133 and 240 nm, as shown in Figure 7b, pointing to a reasonably narrow but not perfectly uniform size distribution across the sample.[55] The potential influence of particle aggregation is observed. Some degree of particle clustering was visible in Figure 7a, though this is generally attributed to surface energy-driven aggregation during specimen preparation rather than any structural bonding between particles.[56] These scale-size particles

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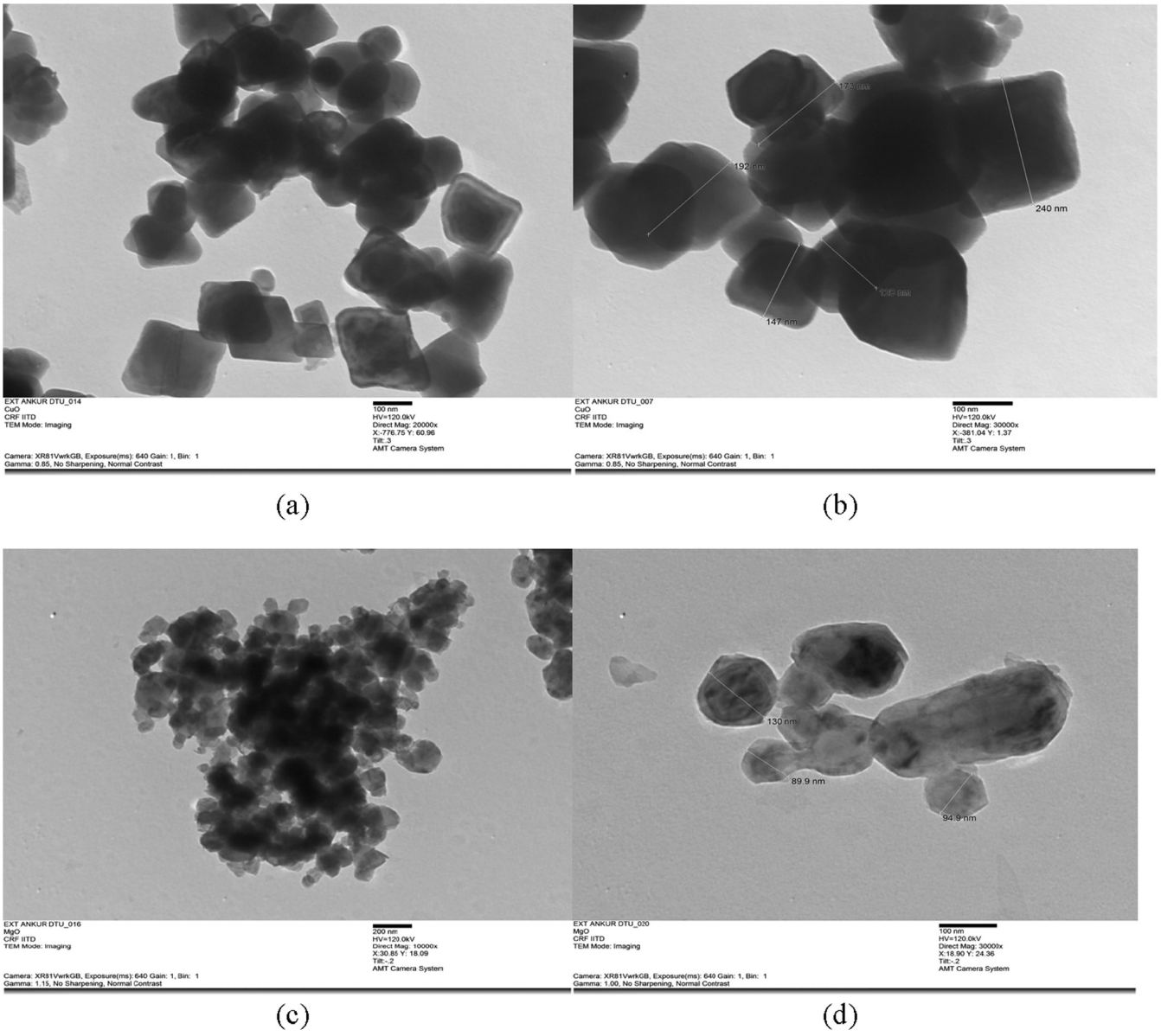


Figure 7. TEM micrographs of CuO (a, b) and MgO (c, d) nanoparticles, showing nonuniform dimensions and particle clustering.

carry a substantial surface-to-volume ratio, which tends to amplify their catalytic behavior inside the combustion chamber.[53] The exposed crystallographic faces of the cubic CuO grains are known to facilitate oxygen-transfer mechanisms, and this is expected to support more thorough fuel oxidation, shorten ignition delay periods, and may contribute to measurable reductions in exhaust pollutants when these nanoparticles are introduced as additives into the test fuel blends.[57]

Following the characterization of CuO nanoparticles, the morphological examination of magnesium oxide (MgO) nanoparticles was similarly conducted using transmission electron microscopy at the same 30,000× magnification and 120 kV accelerating voltage, as shown in Figure 7c,d. Rather than the sharply defined cubic geometry seen in the CuO sample, the MgO nanoparticles displayed a more rounded, quasi-spherical to slightly elongated morphology, with particle boundaries appearing softer and less angular under the electron beam.[58] Size measurements in Figure 7d annotated directly on the micrograph recorded values of approximately 89.9, 130, and 94.9 nm across different particles within the imaged cluster, although overlapping of particles in the micrograph is a limitation, placing the overall size distribution noticeably smaller than that of the CuO fine particles characterized earlier. This reduced particle size is practically significant; smaller nanoparticles inherently offer greater specific surface area, which in the context of fuel-additive applications is associated with enhanced interaction with fuel molecules during the combustion process.[34,59]

From a combustion chemistry standpoint, MgO nanoscale fine particles are well regarded for their ability to act as oxygen carriers and surface reaction promoters within the cylinder environment.[60] Their smaller dimensions in this sample suggest that they would disperse more readily within the fuel matrix compared with CuO, potentially offering more uniform catalytic activity throughout the combustion cycle.[61] When considered alongside the CuO fine particles, the MgO sample adds a complementary dimension to the particle additive strategy employed in this study, one that combines the stronger catalytic oxidation character of CuO with the dispersibility and surface activity advantages offered by the finer MgO particles.[34,53,58]

2.4. Experimental Setup, Procedure, and Engine Specifications

Experimental trials as per the design of experiments (DoE) through CCD-RSM modeling were conducted on a diesel research engine (CRDI) common-rail direct-injection type. Table 5 shows the configuration of the CRDI-VCR engine used. The research engine

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Table 5. Configurations of CRDI Research Diesel Engine Experimental Setup

technical parameters	description
engine	Kirloskar common-rail direct-injection diesel engine, 4 stroke and water cooled
no. of cylinders	1 (single cylinder)
compression ratio	from 12 to 18
cylinder dimensions	87.5 mm of bore and 110 mm of stroke length
combustion chamber volume	661 cm^3
power rating	3.5 kW at 1500 rpm
dynamometer	eddy-current dynamo with a loading unit, water cooled
fuel injector	solenoid based
cooling system rotameters	around 40 LPH to 400LPH, Eureka make
software used	EngineSoft 24.0
temperature sensors	Radix make thermocouple K-type RTD

was operated at multiple engine loads varying from 20% load application to full load with an intermediate range of 40, 60, and 80% of full load conditions, fuel injection pressures (FIPs) from 300 to 700 bar at intervals of 100 bar, with varying HAPO blend ratios, and along with varying numbers of additive particles in blends. Each trial of experiments for different input parameters was conducted after achieving the steady state of engine run with measurements taken after 10−12 min for making the research engine thermally stable and also for the consumption of leftover previous blend oil in the engine setup fuel line after changing the blend ratio.

2.4.1. Engine Instrumentation and Measurements. Engine performance parameters were acquired and monitored from Engine test software (EngineSoft 24.0) provided by APEX Innovations Pvt. Ltd., including brake thermal efficiency (BTHE) and exhaust gas temperature (EGT) using a K-type thermocouple as mentioned in Table 5. Emission parameters acquired from engine software included carbon monoxide (CO), hydrocarbon (HC), and nitrogen oxide (NO_x) emissions by calibrated 5 gas analyzers coupled with it. Before each test sequence, the gas analyzer was reset to avoid measurement errors. Smoke opacity data were acquired using a smoke measurement device attached to the research engine experimental setup. Each output response for CCD experiment was evaluated through three repeated readings after the steady-state condition of running operation is achieved, and the arithmetic mean value was used for RSM modeling in Table 8.

Figure 8 shows the entire experimental setup with a CRDI diesel research engine coupled with the water-cooled eddy-current dynamometer for applying load on the engine. Figure 8 also shows the control panel for varying the engine load and other input parameters with display, rotameters for volumetric flow measurements and other required entities in setup. Figure 9 shows a schematic diagram of the experimental procedure of the test conducted and data acquired through engine software for analyzing performance and emission output responses.

2.5. Measurement Uncertainty Analysis

There are many factors that contributed to the variations and uncertainty in the measurements during experimentation. The techniques involved in measuring data, surrounding environmental condition, and calibration inaccuracy of the experimental setup, all result in uncertainties of the data recorded.[62] For a reliable data assurance, uncertainty analysis has to be done for resolving the errors. Test engine conditions were allowed to stabilize for 10−12 min before recording any output value. For each CCD experimental condition, three repeated steady-state readings were recorded for the measured output response, and the arithmetic mean was used in the RSM data set. The mean square root method is used for combined uncertainty calculation.[63] Percentage errors for all parameters are given in Table 6. The combined uncertainty of the experiment resulted in about 3.98%. Equation 1 used to calculate the combined uncertainty of different parameters is given below.

$$\begin{aligned}&\text{combined uncertainty}\\&= \pm \sqrt{\{(\text{load error})^2 + (\text{temp. sensor error})^2 + (\text{BTE error})^2}\\&\quad + (\text{smoke error})^2 + (\text{CO error})^2 + (\text{NO}_x \text{ error})^2\\&\quad + (\text{uHC error})^2\}\\&= \pm \sqrt{\{(1.78)^2 + (1.45)^2 + (1.15)^2 + (0.9)^2 + (0.4)^2}\\&\quad + (2.57)^2 + (1.3)^2\}\\&= \pm 3.98\% \end{aligned} \quad (1)$$

2.6. CCD-RSM Experimental Design Implementation

Response surface methodology as a statistical tool is used to design the experiments and optimize the results obtained.[64] In RSM, the

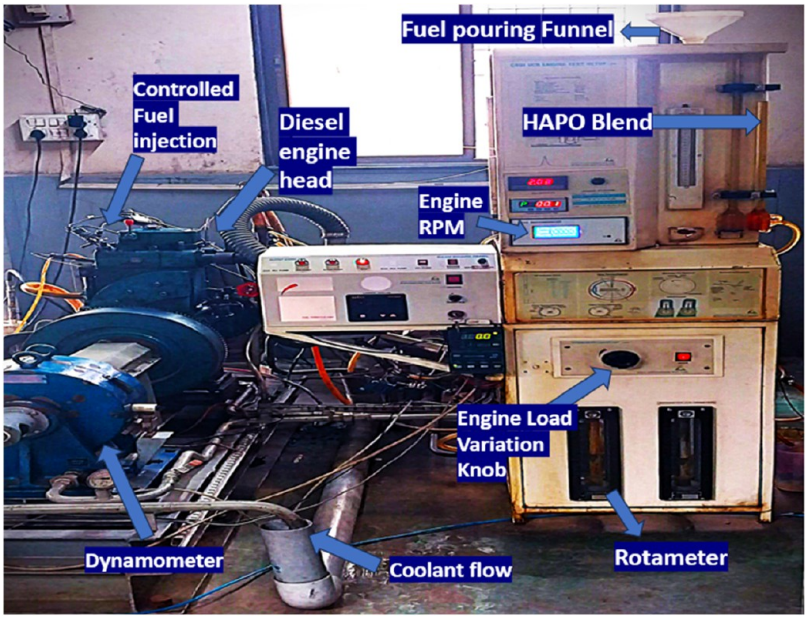


Figure 8. Research engine experimental setup, 1-C, 4-S, CRDI Diesel engine testing facility.

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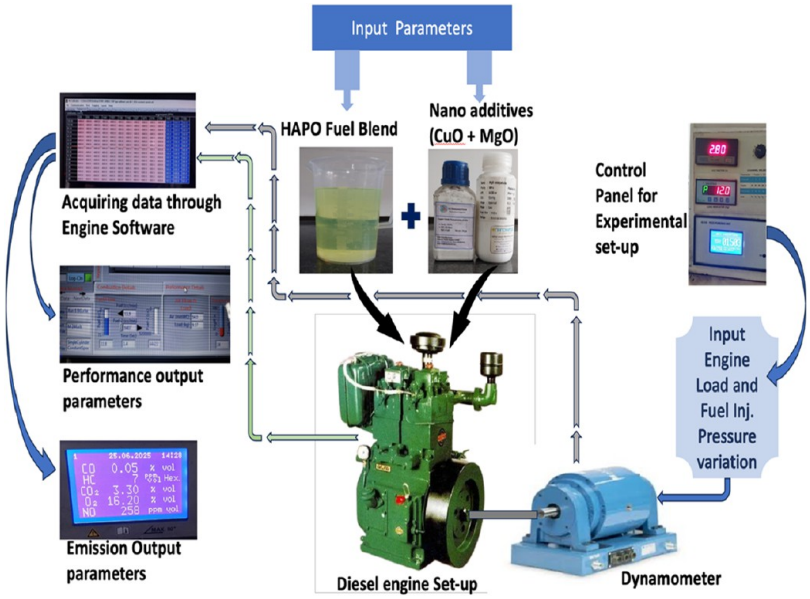


Figure 9. Schematic diagram of the experimental procedure and workflow on CRDI diesel research engine (*Image created by MS-PowerPoint*).

Table 6. Percentage Uncertainty of Different Variables from Engine and Gas Analyzers

variables	uncertainty in %
applied load	±1.78
temperature sensors (EGT)	±1.45
BTHE	±1.15
smoke opacity	±0.9
carbon monoxide (CO)	±0.4
nitrogen oxides (NO_x)	±2.57
unburned hydrocarbons (uHC)	±1.3

central composite design (CCD) approach with 31 experimental data points was used to investigate the effects of four independent variables, i.e., HAPO blend ratio in vol %, nanoparticle concentration (NP) in ppm, fuel injection pressure (FIP) in bar, and engine load in %. The dependent output response parameters considered were BTHE and EGT for performance assessment, and CO, HC, NO_x emissions, and smoke opacity for emission assessment.

Factorial points of the complete design model are taken as three points in the CCD approach.[65] The factorial levels are like an integer number line, where 0 is marked as the center point of the design model to 2 negatives and 2 positive points as −2, −1, 0, +1, +2, shown in the design of experiment in Table 7.

Table 8 shows the design of experiment with these 31 trials or runs of input control variables and the output responses from each trial. These are the observations and readings as a result of the controlled input variables.

Table 7. RSM-CCD Attribute Table for Input Parameters and Their Factorial Levels

	factorial levels				
input parameters	−2	−1	0	+1	+2
HAPO blend %	0	10	20	30	40
NP (CuO + MgO) concentration (ppm)	0	75	150	225	300
FIP (bar)	300	400	500	600	700
engine load (%)	20	40	60	80	100

3. STATISTICAL ANALYSIS AND OPTIMIZATION METHODOLOGY

3.1. RSM Quadratic Model Development

Quadratic equation models of the second order were developed for each response function by the MINITAB 21 software. The design of experiments (DoE) for the analysis was created through the factorial levels as per Table 7, and the number of experimental trials to be conducted has been determined in the CCD-RSM approach[66] by using eq 2.

$$T = 2^n + 2n + k \quad (2)$$

where T is the number of trials to be conducted in the experiment, n signifies the number of input factors, and k is the number of center points. This equation is the combination of factorial points (2^n), axial points ($2n$), and center points (k). Putting the value of n and k in the above equation as 4 and 7, respectively, 4 being the number of input parameters and 7 center values, the number of experimental trials comes out to be 31, shown in Table 6.[23,63]

$$T = (2^4 + 2 \times 4 + 7) = 31 \text{ trials}$$

Relevance tests were conducted for the performance and emission output response parameters by analysis of variance (ANOVA) to observe the influence and significance of the input parameters on the output responses with their interactions. The generated model can be evaluated by assessing the variance fit, i.e., the R^2 coefficient, indicating the accuracy of fitness of data alignment with the model generated. If the determined R^2 values are close to 1, then it indicates a perfect model fit. The optimization is targeted to maximize the performance and minimize the emissions from the engine. The objective function was set to maximize BTHE while minimizing CO, NO_x, smoke, and HC emissions. Then experimental verification of the optimized conditions and respective output response parameters was executed.[67,68]

MINITAB software Version 21 used the regression models to analyze and estimate the output response variables, which are shown by the help of model equations. Performance parameters BTHE and EGT are shown in eqs 3 and 4,

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Table 8. Design of Experiments (DoE) Matrix through RSM-CCD

exp. no.	HAPO blend (%)	NP additives (CuO + MgO) (ppm)	FIP (bar)	load (%)	BTHE (%)	EGT (K)	NO_x (ppmv)	CO (% vol)	HC (ppm)	smoke (%)
1	0	150	500	60	26.15544	553.39	445	0.13893	36	28.74
2	10	75	600	40	16.34977	534.469	335	0.15541	29	19.07175
3	10	225	400	40	17.0226	478.8925	189	0.13996	27	19.63217
4	10	225	400	80	28.09672	569.434	577	0.10597	36	29.27836
5	10	75	400	80	26.79518	637.1695	716	0.13996	46	29.04571
6	10	225	600	80	27.08196	622.7215	828	0.08949	29	29.35384
7	10	225	600	40	16.90127	497.3935	272	0.1173	22	19.67664
8	10	75	600	80	27.9423	700.873	943	0.1173	38	29.12119
9	10	75	400	40	16.17329	514.54	236	0.18528	36	19.03349
10	20	300	500	60	23.27661	496.8265	326	0.09155	21	20.7096
11	20	150	500	60	23.43103	530.1325	440	0.0967	28	26.2353
12	20	150	500	60	23.43103	530.1325	440	0.0967	28	26.2353
13	20	150	300	60	22.65893	499.147	352	0.1379	37	21.77255
14	20	150	500	60	23.43103	530.1325	440	0.0967	28	26.2353
15	20	150	500	100	27.90921	726.283	1050	0.10803	37	26.17119
16	20	150	500	60	23.43103	530.1325	440	0.0967	28	26.2353
17	20	150	500	60	23.43103	530.1325	440	0.0967	28	26.2353
18	20	150	700	60	25.00832	543.142	692	0.0967	28	22.90582
19	20	150	500	60	23.43103	530.1325	440	0.0967	28	26.2353
20	20	150	500	20	10.36	474.42	98	0.1042	16	10.347
21	20	150	500	60	23.43103	530.1325	440	0.0967	28	26.2353
22	20	0	500	60	22.37215	577.918	604	0.13687	33	21.63296
23	30	75	400	80	27.23638	583.252	762	0.11112	37	23.12916
24	30	75	400	40	17.25423	492.259	256	0.14717	28	15.73089
25	30	225	600	80	26.44222	572.7625	881	0.07095	23	22.01968
26	30	225	600	40	17.26526	476.8345	296	0.09361	17	15.01847
27	30	225	400	80	27.20329	527.4655	614	0.08434	28	22.01968
28	30	75	600	80	28.19599	641.2855	959	0.09361	30	23.18603
29	30	225	400	40	16.9123	458.071	205	0.11112	22	14.98745
30	30	75	600	40	16.85715	512.083	360	0.12348	23	15.76398
31	40	150	500	60	22.58172	513.427	555	0.08125	20	21.047

respectively; also, the four emission parameters, CO, smoke, HC, and NO_x, are shown by eqs 5−8, respectively.

$$\begin{aligned}\text{BTE (\%)} = {} & -6.85 + 0.005\ \text{blend (\%)} + 0.0403\ \text{NP (ppm)} + 0.0030\ \text{FIP (bar)} + 0.6218\ \text{load (\%)} + 0.00162\ \text{blend (\%)} \times \text{blend (\%)} \\ & - 0.000040\ \text{NP (ppm)} \times \text{NP (ppm)} + 0.000003\ \text{FIP (bar)} \times \text{FIP (bar)} \\ & - 0.002865\ \text{load (\%)} \times \text{load (\%)} - 0.000297\ \text{blend (\%)} \times \text{NP (ppm)} - 0.000002\ \text{blend (\%)} \times \text{FIP (bar)} - 0.00084\ \text{blend (\%)} \\ & \times \text{load (\%)} - 0.000029\ \text{NP (ppm)} \times \text{FIP (bar)} - 0.000117\ \text{NP (ppm)} \times \text{load (\%)} + 0.000010\ \text{FIP (bar)} \times \text{load (\%)} \end{aligned} \quad (3)$$

$$\begin{aligned}\text{EGT (K)} = {} & 512.1 + 0.26\ \text{blend (\%)} - 0.058\ \text{NP (ppm)} + 0.113\ \text{FIP (bar)} - 3.212\ \text{load (\%)} \\ & + 0.0134\ \text{blend (\%)} \times \text{blend (\%)} + 0.000415\ \text{NP (ppm)} \times \text{NP (ppm)} - 0.000172\ \text{FIP (bar)} \times \text{FIP (bar)} + 0.0452\ \text{load (\%)} \times \text{load (\%)} \\ & + 0.00207\ \text{blend (\%)} \times \text{NP (ppm)} - 0.00084\ \text{blend (\%)} \times \text{FIP (bar)} - 0.0373\ \text{blend (\%)} \times \text{load (\%)} - 0.000214\ \text{NP (ppm)} \\ & \times \text{FIP (bar)} - 0.00533\ \text{NP (ppm)} \times \text{load (\%)} + 0.00448\ \text{FIP (bar)} \times \text{load (\%)} \end{aligned} \quad (4)$$

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Table 9. ANOVA Results of Performance Parameters: BTHE, EGT, and Smoke

	BTHE (%)			EGT(K)			smoke opacity (%)		
source	adj SS	*F*-value	*P*-value	adj SS	*F*-value	*P*-value	adj SS	*F*-value	*P*-value
model	639.769	54.80	0.000	119,715	111.45	0.000	724.327	221.90	0.000
linear	596.370	178.79	0.000	106,673	347.59	0.000	555.592	595.74	0.000
blend (%)	1.573	1.89	0.189	5748	74.91	0.000	138.931	595.88	0.000
NP (ppm)	0.155	0.19	0.672	13,754	179.27	0.000	0.648	2.78	0.115
FIP (bar)	1.059	1.27	0.276	6187	80.64	0.000	0.286	1.23	0.284
load (%)	593.583	711.84	0.000	80,985	1055.56	0.000	415.727	1783.06	0.000
square	40.921	12.27	0.000	9751	31.77	0.000	159.811	171.36	0.000
blend (%) × blend (%)	0.754	0.90	0.356	52	0.67	0.424	3.335	14.31	0.002
NP (ppm) × NP (ppm)	1.431	1.72	0.209	156	2.03	0.173	46.274	198.47	0.000
FIP (bar) × FIP (bar)	0.023	0.03	0.869	85	1.11	0.309	27.469	117.82	0.000
load (%) × load (%)	37.564	45.05	0.000	9347	121.83	0.000	114.398	490.65	0.000
2-way interaction	2.479	0.50	0.803	3290	7.15	0.001	8.924	6.38	0.001
blend (%) × NP (ppm)	0.793	0.95	0.344	39	0.50	0.488	1.845	7.91	0.012
blend (%) × FIP (bar)	0.000	0.00	0.993	11	0.15	0.705	0.001	0.00	0.954
blend (%) × load (%)	0.449	0.54	0.474	891	11.61	0.004	6.932	29.73	0.000
NP (ppm) × FIP (bar)	0.735	0.88	0.362	41	0.54	0.475	0.000	0.00	0.979
NP (ppm) × load (%)	0.494	0.59	0.453	1025	13.35	0.002	0.145	0.62	0.441
FIP (bar) × load (%)	0.006	0.01	0.931	1284	16.73	0.001	0.000	0.00	0.975

$$\begin{aligned}\text{CO (\% vol)} = {} & 0.5411 - 0.00466\ \text{blend (\%)} - 0.000725\ \text{NP (ppm)} - 0.000815\ \text{FIP (bar)} \\ & - 0.00233\ \text{load (\%)} + 0.000043\ \text{blend (\%)} \times \text{blend (\%)} + 0.000001\ \text{NP (ppm)} \times \text{NP (ppm)} \\ & + 0.000001\ \text{FIP (bar)} \times \text{FIP (bar)} + 0.000008\ \text{load (\%)} \times \text{load (\%)} \\ & + 0.000002\ \text{blend (\%)} \times \text{NP (ppm)} + 0.000001\ \text{blend (\%)} \times \text{FIP (bar)} \\ & + 0.000009\ \text{blend (\%)} \times \text{load (\%)} + 0.000000\ \text{NP (ppm)} \times \text{FIP (bar)} \\ & + 0.000002\ \text{NP (ppm)} \times \text{load (\%)} + 0.000001\ \text{FIP (bar)} \times \text{load (\%)} \end{aligned} \quad (5)$$

$$\begin{aligned}\text{smoke (\%)} = {} & -36.54 + 0.1649\ \text{blend (\%)} + 0.0787\ \text{NP (ppm)} + 0.0992\ \text{FIP (bar)} + 0.8825\ \text{load (\%)} \\ & - 0.003415\ \text{blend (\%)} \times \text{blend (\%)} - 0.000226\ \text{NP (ppm)} \times \text{NP (ppm)} \\ & - 0.000098\ \text{FIP (bar)} \times \text{FIP (bar)} - 0.005000\ \text{load (\%)} \times \text{load (\%)} \\ & - 0.000453\ \text{blend (\%)} \times \text{NP (ppm)} - 0.000007\ \text{blend (\%)} \times \text{FIP (bar)} \\ & - 0.003291\ \text{blend (\%)} \times \text{load (\%)} - 0.000000\ \text{NP (ppm)} \times \text{FIP (bar)} \\ & - 0.000064\ \text{NP (ppm)} \times \text{load (\%)} + 0.000002\ \text{FIP (bar)} \times \text{load (\%)} \end{aligned} \quad (6)$$

$$\begin{aligned}\text{HC (ppm)} = {} & 70.92 - 0.572\ \text{blend (\%)} - 0.0612\ \text{NP (ppm)} - 0.01572\ \text{FIP (bar)} + 0.473\ \text{load (\%)} \\ & + 0.00191\ \text{blend (\%)} \times \text{blend (\%)} - 0.000015\ \text{NP (ppm)} \times \text{NP (ppm)} \\ & + 0.000126\ \text{FIP (bar)} \times \text{FIP (bar)} - 0.000565\ \text{load (\%)} \times \text{load (\%)} \\ & + 0.000651\ \text{blend (\%)} \times \text{NP (ppm)} + 0.000366\ \text{blend (\%)} \times \text{FIP (bar)} \\ & - 0.00244\ \text{blend (\%)} \times \text{load (\%)} + 0.000049\ \text{NP (ppm)} \times \text{FIP (bar)} \\ & - 0.000325\ \text{NP (ppm)} \times \text{load (\%)} - 0.000183\ \text{FIP (bar)} \times \text{load (\%)} \end{aligned} \quad (7)$$

$$\begin{aligned}\text{NO}_x\ \text{(ppm)} = {} & 817 - 6.32\ \text{blend (\%)} - 0.806\ \text{NP (ppm)} - 2.538\ \text{FIP (bar)} - 5.67\ \text{load (\%)} \\ & + 0.1683\ \text{blend (\%)} \times \text{blend (\%)} + 0.001483\ \text{NP (ppm)} \times \text{NP (ppm)} \\ & + 0.002233\ \text{FIP (bar)} \times \text{FIP (bar)} + 0.0883\ \text{load (\%)} \times \text{load (\%)} + 0.00185\ \text{blend (\%)} \times \text{NP (ppm)} \\ & - 0.00006\ \text{blend (\%)} \times \text{FIP (bar)} + 0.0208\ \text{blend (\%)} \times \text{load (\%)} + 0.000538\ \text{NP (ppm)} \times \text{FIP (bar)} \\ & - 0.01056\ \text{NP (ppm)} \times \text{load (\%)} + 0.01762\ \text{FIP (bar)} \times \text{load (\%)} \end{aligned} \quad (8)$$

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Table 10. ANOVA Results of Emission Parameters (CO, NO_x, HC)

source	HC (ppmv)			NO_x (ppmv)			CO (% vol)		
	adj SS	*F*-value	*P*-value	adj SS	*F*-value	*P*-value	adj SS	*F*-value	*P*-value
model	1320.90	59.85	0.000	1,832,935	244.03	0.000	0.017321	10.34	0.000
linear	1252.65	198.65	0.000	1,759,013	819.65	0.000	0.014885	31.10	0.000
blend (%)	312.89	198.48	0.000	8744	16.30	0.001	0.004555	38.07	0.000
NP (ppm)	307.36	194.98	0.000	66,550	124.04	0.000	0.005140	42.96	0.000
FIP (bar)	183.53	116.42	0.000	166,298	309.96	0.000	0.002525	21.10	0.000
load (%)	448.87	284.74	0.000	1,517,420	2828.28	0.000	0.002666	22.28	0.000
square	50.39	7.99	0.001	49,460	23.05	0.000	0.002139	4.47	0.013
blend (%) × blend (%)	1.05	0.66	0.427	8100	15.10	0.001	0.000538	4.50	0.050
NP (ppm) × NP (ppm)	0.20	0.12	0.729	1870	3.48	0.080	0.000824	6.89	0.018
FIP (bar) × FIP (bar)	45.34	28.76	0.000	14,259	26.58	0.000	0.001079	9.02	0.008
load (%) × load (%)	1.46	0.93	0.350	35,695	66.53	0.000	0.000320	2.67	0.121
2-way interaction	17.86	1.89	0.145	24,462	7.60	0.001	0.000296	0.41	0.860
blend (%) × NP (ppm)	3.81	2.42	0.140	31	0.06	0.814	0.000056	0.47	0.505
blend (%) × FIP (bar)	2.14	1.36	0.261	0	0.00	0.992	0.000024	0.20	0.661
blend (%) × load (%)	3.81	2.42	0.140	277	0.52	0.483	0.000056	0.47	0.505
NP (ppm) × FIP (bar)	2.14	1.36	0.261	261	0.49	0.496	0.000035	0.29	0.596
NP (ppm) × load (%)	3.81	2.42	0.140	4017	7.49	0.015	0.000091	0.76	0.397
FIP (bar) × load (%)	2.14	1.36	0.261	19,877	37.05	0.000	0.000035	0.29	0.596

Table 11. Regression Analysis of Performance and Emission Output Responses

output response	residual standard deviation	R^2	R^2 (adjusted)	R^2 (predicted)	model assessment
BTHE (%)	0.913169	97.96%	96.17%	88.23%	satisfactory
EGT (K)	8.75915	98.98%	98.10%	94.15%	satisfactory
smoke opacity (%)	0.482860	99.49%	99.04%	97.05%	satisfactory
HC (ppmv)	1.25556	98.13%	96.49%	89.21%	satisfactory
NO_x (ppmv)	23.1628	99.53%	99.13%	97.31%	satisfactory
CO (% vol)	0.0109384	90.05%	81.34%	42.68%	limited predictive capability

3.2. Analysis of Variance, Model Accuracy, and Residual Diagnostics

Relevance test evaluation for the engine performance and emission characteristics can be conducted by ANOVA (analysis of variance). Tables 9 and 10 show the ANOVA results of performance and emission output responses. The measured values of responses are compared to check whether the investigated variables have any significant influence on the responses.[69]

The ANOVA results indicate that the engine load and fuel injection pressure significantly influenced all response parameters. The R^2 values for the developed models were: BTHE (0.98), EGT (0.99), CO (0.9), HC (0.98), NO_x (0.99), and smoke (0.99), as shown in Table 11. The response models were evaluated by using R^2, adjusted R^2, and predicted R^2. The CO model showed an R^2 value of 90.05% and an adjusted R^2 value of 81.34%, indicating a satisfactory fit to the experimental observations. However, its predicted R^2 value was 42.68%, indicating limited predictive capability for new observations. Therefore, the CO model is comparatively less reliable as an optimization component than other responses and was used cautiously for trend interpretation and multiresponse optimization within the experimental design space. CO-related optimization results are therefore treated as exploratory and are not presented as independently validated predictive conclusions.[70]

The ANOVA *P*-values showed that engine load was the most significant factor affecting engine performance and emissions, with *P*-values less than 0.001 for most responses.[71] The addition of powdered particles showed a significant effect on NO_x reduction ($P < 0.01$) and moderate effects on CO and HC emissions. Fuel injection pressure had pronounced effects on BTHE and EGT, while the HAPO blend ratio showed interactive effects with the other variables.

In statistical analysis, after conducting an ANOVA test to evaluate the accuracy of the model, the residual plots for each output response are examined. The residual plots are generated to depict the differences between data predicted through the model and data observed experimentally.[25] For each data set, these differences in predicted and observed data result in "residuals." Normal probability graphs of residuals for all output responses, i.e., BTHE, EGT, HC, smoke, CO, and NO_x, are shown in Figure 10 in order. These plots show that all residuals nearly follow a linear straight line, which is an indication that they are distributed normally.[72] It also depicts that errors within the experiments are random and do not follow a systematic pattern. Data points alignment within the plot's reference line assures that the statistical model used with quadratic equation for prediction is valid.[12] Minimum deviations of the data points from the line confirm less error in the statistical model, which supports the reliability factors of the results in experimentation.[21,63]

4. RESULTS AND DISCUSSION

The interaction effects among various input variables with the output responses were analyzed using the surface plots and contours generated by response surface methodology. Surface plots are three-dimensional figures, as represented in Figures 11−16, that depict the relationship in a nonlinear way.[73]

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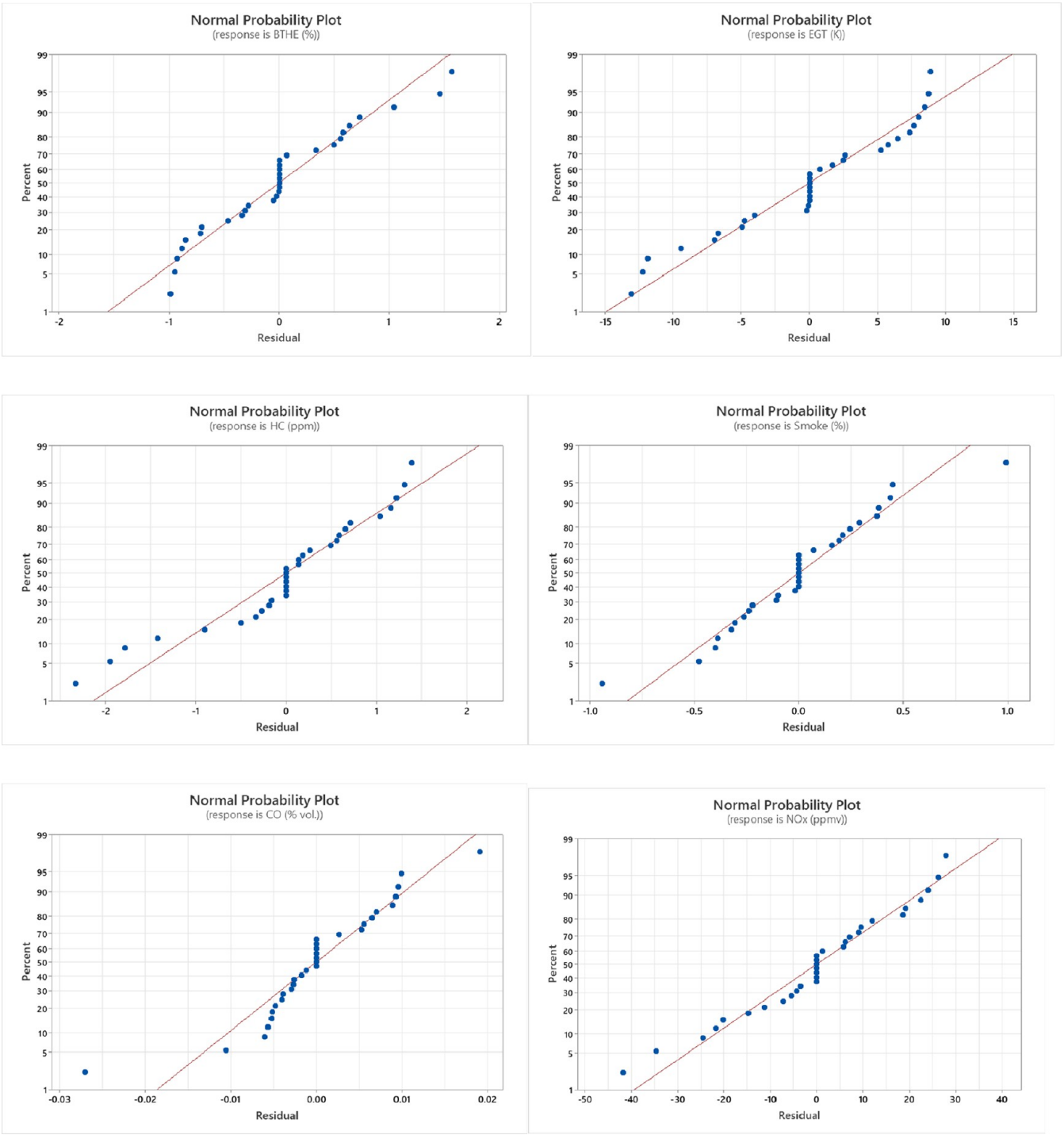


Figure 10. Normal probability plots of residuals for quadratic RSM models of output responses.

4.1. Performance Analysis of HAPO Blends

4.1.1. Brake Thermal Efficiency (BTHE). The energy conversion ratio of fuel is indicated by brake thermal efficiency, which shows how much chemical energy of the fuel is converted into mechanical energy or mechanical work. Thus, brake thermal efficiency represents the prominent performance characteristics within the engine parameters. Results of the experiments show BTHE has significant variations for different input parameters. Figure 11 shows the three-axis response surface graphs representing the combined effect of all four varied input conditions in three plots.

Figure 11a shows the effect of two novel input conditions for this research, i.e., HAPO blend in percentage and the concentration of metal oxide (CuO and MgO) nanoparticles as fuel additives on the performance output response parameter BTHE, keeping the other two input parameters on hold as constant at 500 bar FIP and engine load to be 60%. BTHE gets reduced with the increase of HAPO blend ratio in diesel fuel because of the combined influence of comparatively lower calorific value, density, and fuel viscosity increment due to the presence of algae oil as oxygenated bio-oil, fuel injection pressure, and engine load impacted it further. However, the

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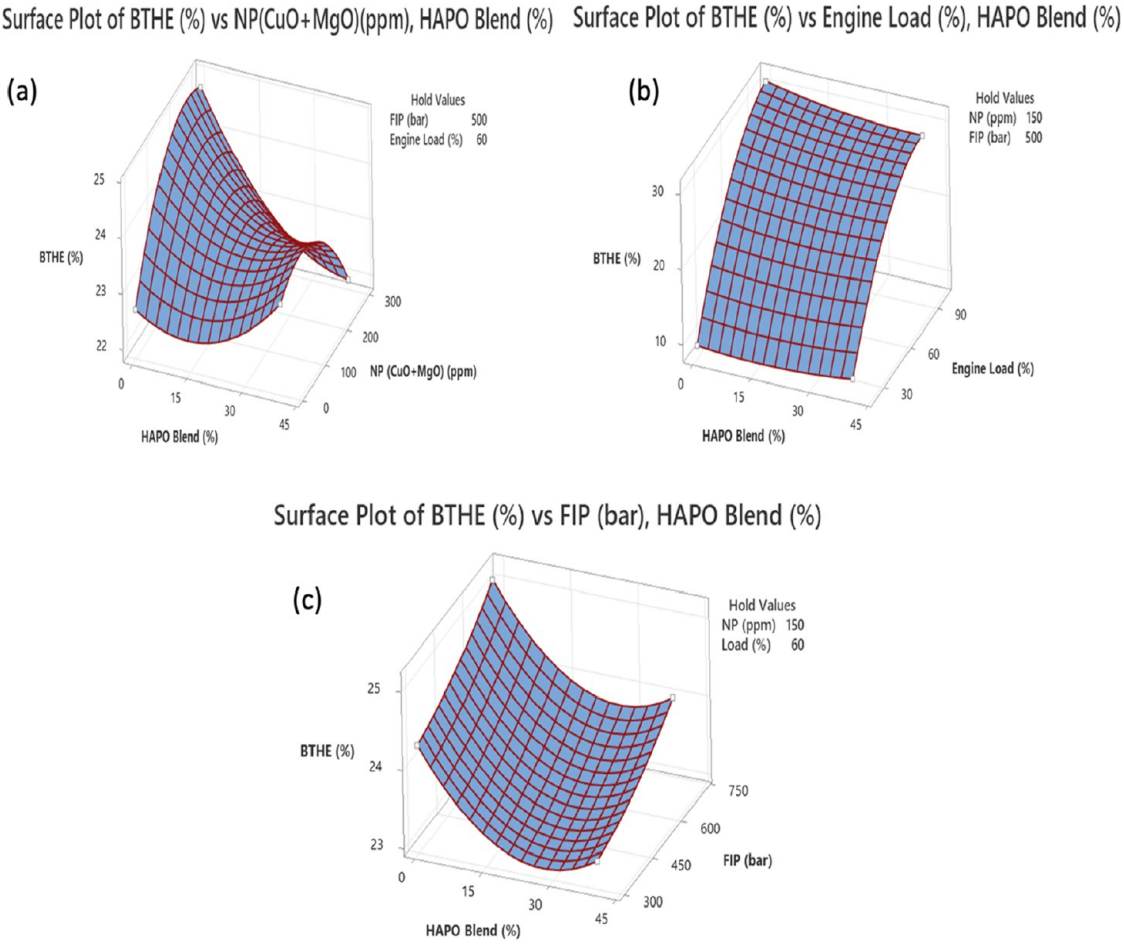


Figure 11. Response surface plots showing effects of (a) HAPO blend and binary additive concentration, (b) HAPO blend and engine load, and (c) HAPO blend and fuel injection pressure on BTHE (%).

addition of plastic oil with the algae has increased it to some extent due to the better heating value of plastic-derived oil than algae oil as plastic itself is a petroleum byproduct.[28] As shown in the graph, inclusion of copper and magnesium oxide nanoparticles has further improved the BTHE at initial concentration levels due to catalytic reactions, which is consistent with improved combustion, but slightly reduces at higher ppm levels due to agglomeration of nanoparticles leading to bad fuel spray characteristics.[74] The combined effect surface plot in Figure 11a also shows that at higher HAPO blend % with the high concentration of these nanoadditives, the elevated trendline of the BTHE has taken a dip, depicting the optimal limit for addition of both the novel inputs HAPO blend % and nanoparticle concentration to the diesel fuel for better performance.

Figure 11b shows the impact of varying engine load with the HAPO blend % on BTHE, keeping the same constant 500 bar FIP and concentration of NP at 150 ppm. Engine load has been varied from low intensity to full, taken in percentage of full load applied through the dynamometer from 20 to 100%. Increasing engine load improves the temperature inside the cylinder, indicating better combustion with less heat rejection, resulting in increased thermal efficiency. The negatives of using HAPO-blended diesel fade a little at higher load on the engine, which indicates better utilization of this blend at heavy-load conditions. Similar observations and results can be seen in the previous research articles.[20,75]

Figure 11c shows the influence of fuel injection pressure (FIP) in bar with the varying blend ratio of HAPO. Here, engine load is fixed at 60% of the full dynamo loading with the same constant NP concentration of 150 ppm. An increase in the injection pressure of fuel up to a certain range will enhance the BTHE due to high atomization of fuel droplets, leading to higher homogeneity of the air and fuel mixture. After a certain range, high fuel pressure may cause more fuel to cling to cylinder walls and subsequently may not take part in combustion. However, in this surface plot, that is not visible.

4.1.2. Exhaust Gas Temperature (EGT). Exhaust gas temperature is considered as an important indicator for assessing combustion efficiency and combustion chamber temperature with fuel consumption. Figure 12 represents the influence of all four input parameters on EGT. Figure 12a shows the effect of HAPO blend % and NP concentration on EGT at constant 500 bar FIP and 60% of full engine load. The temperature (EGT) has reduced moderately from around 625 to 550 K by increasing the total binary (CuO + MgO) concentration from 0 to 300 ppm. This indicates that more heat conversion into mechanical work may enhance the combustion efficiency due to catalytic reactions of metal oxide nanoparticle addition. Copper and magnesium oxides may support rapid oxidation and better combustion, reducing the heat rejection and thus lowering the EGT. The smooth declining curve indicates a significant effect of nanoparticle addition. By increasing the HAPO blend ratio in neat diesel

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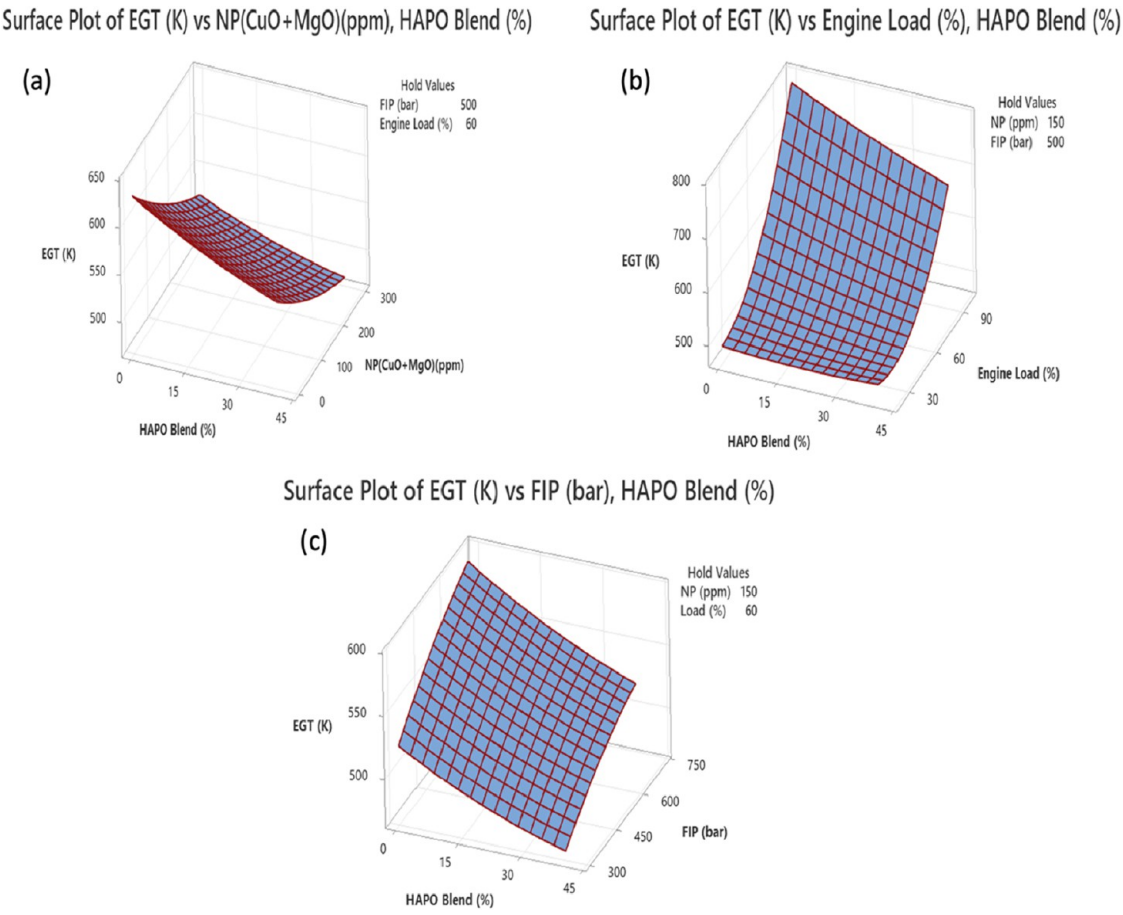


Figure 12. Response surface plots showing effects of (a) HAPO blend and binary additive concentration, (b) HAPO blend and engine load, and (c) HAPO blend and fuel injection pressure on EGT (K).

from 0 to 40 vol %, a marginal reduction in EGT has been observed around 20−25 K due to a similar reason for lower calorific value, and also oxygenated algae bio-oil might absorb more heat in combustion as it has high latent heat of vaporization, which was also observed in the previous research.[76]

Figure 12b shows that exhaust temperature sharply increases with an increase in engine load, from 480 to 500 K at lower load % to about 790−810 K at higher loads without any impact of blending percentage; but at a higher HAPO blend ratio, EGT behavior is dominated by the higher load. This signifies a weak interaction between both input parameters for EGT, as the steep slope of the engine load axis in the plot shows dominance affecting EGT.

Figure 12c suggests a positive correlation of FIP with EGT; by increasing fuel injection pressure from 300 to 700 bar, exhaust gas temperature also increases due to better atomization of fuel and its mixing with air, leading to high heat release, but the impact of higher blend ratio has gradually reduced the EGT, which may be because of more heat consumption during combustion of oxygenated algae bio-oil.[77,78]

4.2. Emission Analysis of HAPO Blends

4.2.1. Carbon Monoxide (CO) Emissions. The impact of all four input variables on carbon monoxide emissions (vol %) as pollutants generally formed from incomplete or partial combustion of fuel is analyzed in Figure 13. CO emissions showed the scripted correlation with engine load, nanoparticle concentration, and fuel injection pressure. The trend of all three surface plots is similar, indicating a strong positive response of these input variables on CO emissions. Increasing load, NP, and FIP has significantly reduced CO emissions initially and slightly elevated at the high values of the input variables. Increasing HAPO blend concentration in diesel from 0 to 30% results in a noticeable reduction of carbon monoxide emissions from 0.185 to 0.08 vol %, as shown in Figure 13a. This reduction is consistent with the presence of inherent oxygen molecules in bioalgal oil, supporting better oxidation, and subsequently may indicate better combustion. However, at higher blending around 40 vol %, this reduction levels off due to viscousness of bioalgal oil, leading to less atomization.[30,79] Addition of copper and magnesium oxide nanoparticles at optimum levels has further reduced the CO emissions from 0.13 to 0.09 vol % at an HAPO blending concentration of 20% due to the catalytic activity of metal oxides, indicating enhanced oxidation reactions and better fuel combustion.

The influence of engine load and HAPO blend is visible in Figure 13b. CO decreases with an increase in load %, from 0.11 vol % at 20% engine load to 0.07 vol % at 80% engine load, keeping the blend ratio at 30% as per the output response shown in Table 8. This happens as the temperature inside the engine cylinder increases at higher load, which might support the combustion reaction.[80,81] Also, proper air-fuel mixing at high values of FIP shows reduced emissions of carbon monoxide in Figure 13c.

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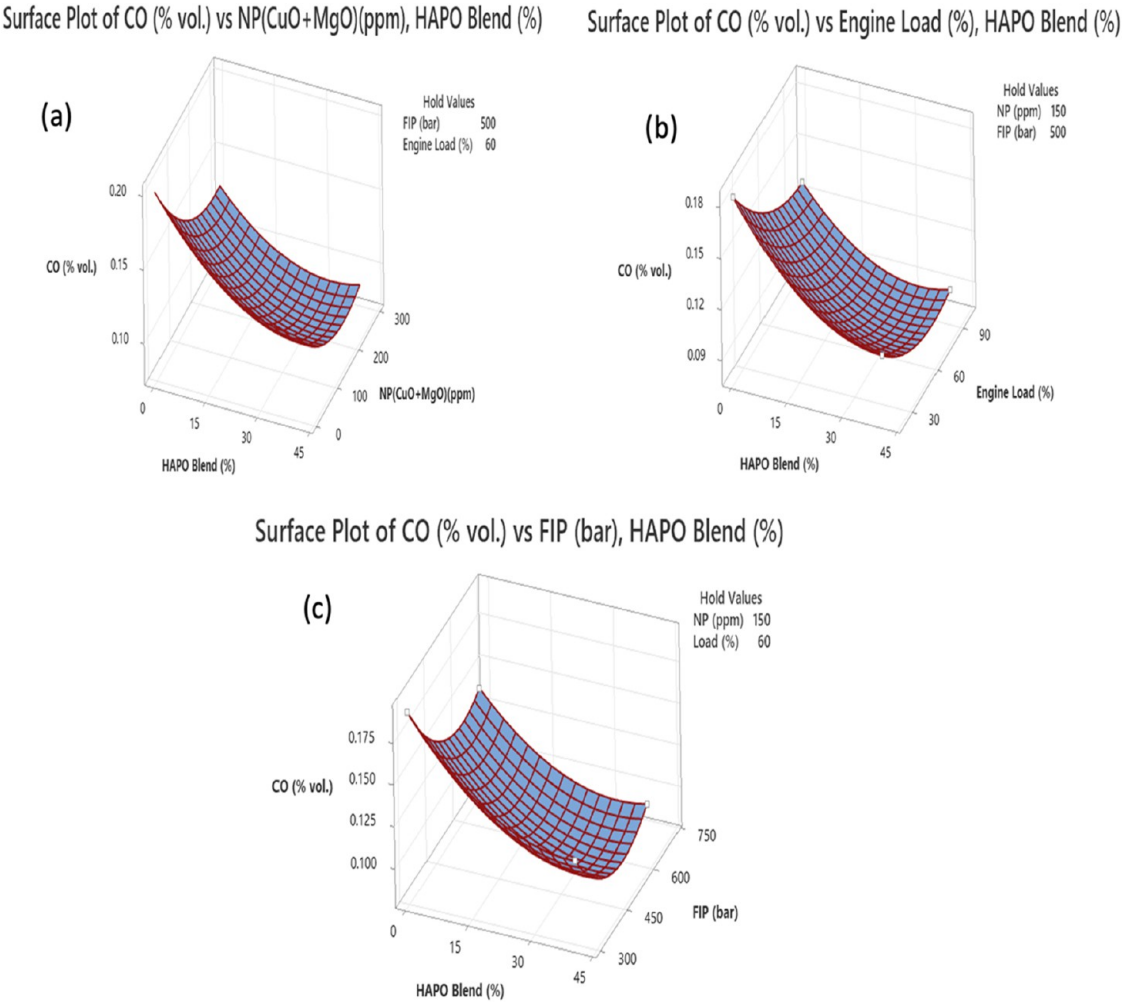


Figure 13. Response surface plots showing effects of (a) HAPO blend and binary additive concentration, (b) HAPO blend and engine load, and (c) HAPO blend and fuel injection pressure on CO emissions (% vol).

4.2.2. Hydrocarbon (HC) Emissions. Unburned hydrocarbon emissions also result normally from incomplete combustion of fuel molecules, just like carbon monoxide, particularly at lower loads when the in-cylinder temperature remains below average combustion temperatures. Also, sometimes the fuel-rich regions within the combustion chamber due to the fuel quenching effect on the cylinder liner may contribute to unburned hydrocarbons.[24,82] Keeping the remaining two input conditions constant, Figure 14 shows the interaction among 4 input variables and their cumulative effect on HC emissions. The response surface plot shows that HC emissions have dropped from 46 to 22 ppm with the increasing amount of NP added at 75−225 ppm, respectively; CuO + MgO nanoparticles may have enhanced the flame propagation through catalytic oxidation, indicating better combustion.[35,36,59] The smooth declining surface in Figure 14a indicates a stronger correlation of these metal oxide nanoadditive concentrations with HC emissions. Additionally, the increase in the HAPO blend ratio also leads to the reduction of hydrocarbon emissions by 16 ppm from using neat diesel to 40 vol % HAPO-blended diesel, which may be due to better hydrocarbon oxidation and flame quenching. Figure 14b shows the strong impact of engine load on the HC emissions; elevated slope of 21 ppm of HC can be seen from 16 to 37 ppm for low load at 20% and the full load condition, respectively, keeping the HAPO blend ratio also at 20%. This elevated steep gradient with the load axis confirms that the load is the most dominating input variable affecting HC emissions. However, this increment of HC can be compensated by increasing the blending ratio and optimum fuel injection pressure, which makes this elevation decline. As shown in Figure 14c, beyond the optimum range of FIP, HC emissions become marginal due to impingement of fuel droplets on the combustion chamber liner, indicating the slight flatness of the surface curve for HC output response, which is correlated with previous studies.[10,15]

4.2.3. Nitrogen Oxide (NO_x) Emissions. The most significant pollutants from the diesel engines are oxides of nitrogen. NO_x is formed by an endothermic interaction between the free nitrogen entering with air and the necessary oxygen molecules inside the cylinder due to high temperature during combustion. Availability of oxygen and reaction time are the key parameters affecting NO_x emissions.[37,83] Figure 15 shows the cumulative influence of HAPO blend, CuO + MgO nanoparticles, FIP, and engine load on NO_x emissions.

The combined effect of varying NP concentration and varying HAPO blends at a constant FIP of 500 bar and 60% load is shown in Figure 15a, presenting a very mild quadratic curve trend with copper and magnesium oxide concentration. Initially, the NO_x emission curve has declined from 604 to 440

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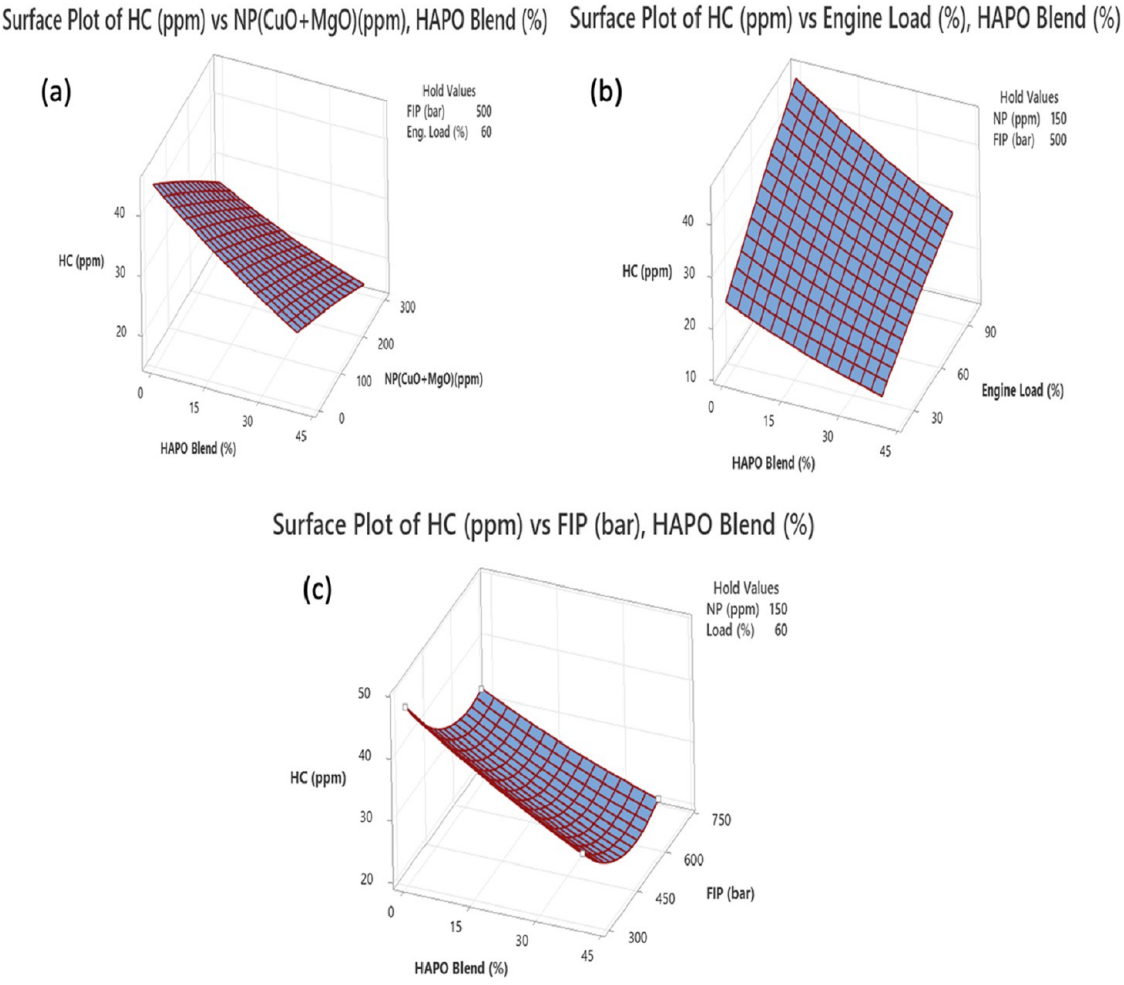


Figure 14. Response surface plots showing effects of (a) HAPO blend and binary additive concentration, (b) HAPO blend and engine load, and (c) HAPO blend and fuel injection pressure on unburned HC emissions.

ppm for nanoparticle concentrations varying from 0 to 150 ppm, respectively, and has further reduced to 346 ppm at a maximum total binary additive concentration of 300 ppm without HAPO blending, indicating a positive impact of metal oxide nanoparticles on reducing NO_x. However, as HAPO blend concentration increases from 0 to 40 vol %, the surface plot of NO_x has shown an upward bend reflecting the increased emission value of NO_x due to the availability of more oxygen molecules of algae oil within the HAPO blend. However, as expected for a biofuel, NO_x has not increased much for the hybrid algae-plastic oil blend comparatively. At the maximum HAPO blending ratio of 40 vol %, keeping all other input parameters at optimally constant, i.e., NP = 150 ppm, FIP = 500 bar, and 60% of engine load, the emission value of NO_x just rose to 555 ppm, which is significantly lower than that of one oxygenated biofuel blend used in the engine. The reason lies in the hybridization of these two oils derived from algae and plastic. As oil derived from waste plastic is itself a petroleum product without the significant presence of oxygen molecules, the overall oxygen concentration at a high blending ratio of HAPO blends becomes low, resulting in comparatively lower NO_x emissions.[63,84,85] Therefore, blending a non-oxygenated waste oil into the biofuel blend can favor possibilities of higher blending percentages in diesel.

Figure 15b shows that until the central value of FIP, increasing the HAPO blend slightly lowers NO_x emissions up to moderate blends, but at higher FIP, this reduction effect becomes insignificant. This indicates that at high injection pressures, the cooling effect of algae oil vanishes. Similarly, in Figure 15c, increasing engine load demonstrated a sharp elevation of NO_x emissions, from 98 to 1050 ppm at 20 and 100% engine loading, respectively, keeping the HAPO blend ratio the same at 20% and the other two factors as optimal constants.

The overall inference reveals that high engine load and higher injection pressures increase NO_x emissions. The moderate NP concentration around 150 ppm and 20% HAPO blend ratio help in controlling NO_x emissions. This trend is consistent with previous studies with greater thermal loading and possibly higher in-cylinder temperatures at higher loads. However, direct temperature or combustion data were not available. Therefore, there is a need for further combustion analysis to support the NO_x formation trend. The current NO_x trends are specific to fuel-additive composition and operating conditions and should be interpreted as empirical results within the investigated domain.

4.2.4. Smoke Opacity. Smoke opacity is associated with particulate matter and soot formation due to incomplete or partial combustion; fuel-rich regions within the combustion chamber contribute to high smoke formation.[86,87] Figure 16 illustrates the combined impact of copper and magnesium oxide nanoparticles, FIP, load, and HAPO blend on the

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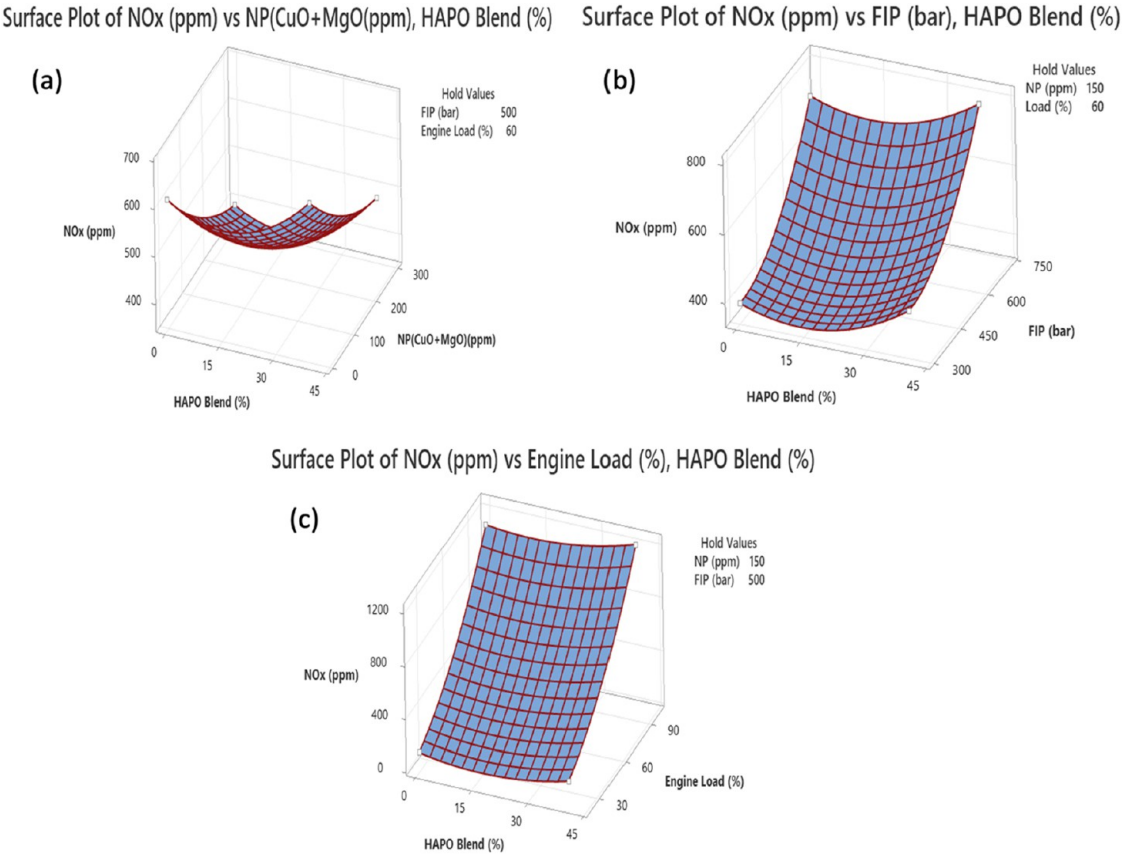


Figure 15. Response surface plots showing effects of (a) HAPO blend and binary additive concentration, (b) HAPO blend and fuel injection pressure, and (c) HAPO blend and engine load on NO_x emissions.

formation of smoke measured in percentage of its opacity. At fixed values of 500 bar FIP and engine load of 60%, Figure 16a shows a noticeable decrement in smoke opacity with an increase in the concentration of CuO and MgO nanoparticles above 200 ppm and an increase in the HAPO blend percentage in neat diesel up to 40 vol %. Gradual increment of HAPO blend ratios in diesel continuously decreases the smoke opacity by 8−10% due to the availability of oxygen molecules that suppress soot formation. While by increasing the NP, there is 6% smoke reduction from 26.23 to 20.7% on doubling the total binary additive concentration from 150 to 300 ppm, respectively, which can be inferred from Table 8 as well. This reduction is the effect of catalytic oxidation reactions of metal oxide nanoparticles enhancing oxidation of soot and particulates, which may be associated with better combustion. Below 150 ppm of NP, smoke formation was higher due to agglomeration of nanoparticles affecting the surface area.[73,75] Similar trends were seen in Figure 16c for the impact of the FIP blend on smoke opacity. At lower injection pressures, smoke opacity was elevating, but after the optimum FIP level of 500 bar is reached, the reducing trend was observed. Smoke opacity got reduced by around 7−9% due to enhanced fuel atomization and homogeneous charge formation. In Figure 16b, the dominance effect among all input parameters is shown by the engine load on smoke opacity. An almost linear elevated steep gradient is observed for engine load affecting the smoke opacity, and the smoke significantly increased from 10.36% at 20% engine load to about 26.17% at full load, reflecting a 16% increase in smoke opacity approximately.

5. DESIRABILITY OPTIMIZATION AND CONFIRMATION OF RESULTS

5.1. Interaction Effects

The interaction effects between the FIP (fuel injection pressure) and load on the engine in Figures 11b,c−16b,c show that an optimal FIP value varies depending upon the engine load. To have maximum brake thermal efficiency (BTHE), an FIP of 650−700 bar is preferred, but for reduction of NO_x emissions, a medium pressure range of around 500−600 bar along with the use of metal oxide nanoadditives is optimum.

The interaction effect between the nanoadditive concentration and engine load in Figures 11−16 of surface plots demonstrates that the benefits of adding nanoparticles in fuel are more at medium to high engine load; when the engine load applied is less, the effects are minimal, and at high engine loads, the benefits of catalysts are maximum.

5.2. Optimized Operating Conditions

The process of optimization maximizes the brake thermal efficiency as a good performance indicator and minimizes the polluting emissions such as smoke, CO, uHC, and NO_x. By RSM modeling and desirability function analysis through MINITAB 21, the optimum values for the operating conditions were finalized within the specified experimental design space. The influence of HAPO blend ratio, FIP, and load is optimized on these output responses. These optimized output responses can be seen in Figure 17.

Figure 18 shows the optimization plot depicting the optimized input operating parameters that resulted from

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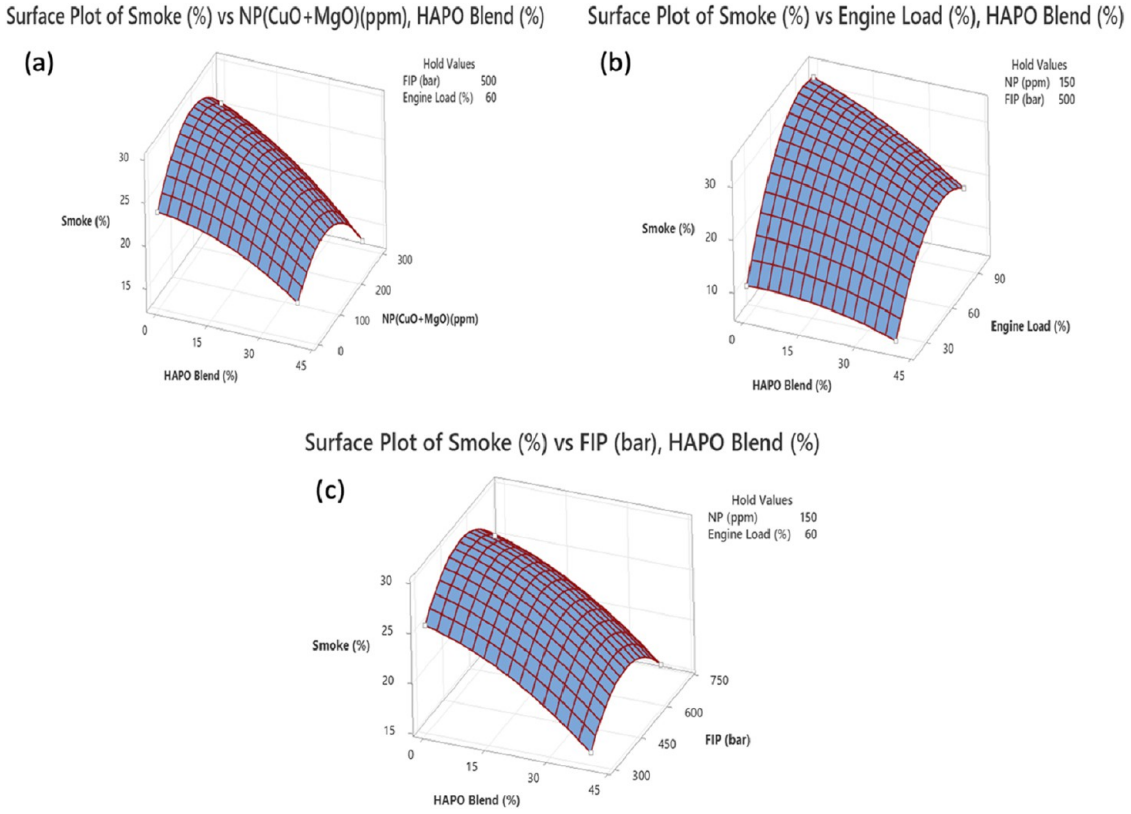


Figure 16. Response surface plots showing effects of (a) HAPO blend and binary additive concentration, (b) HAPO blend and engine load, and (c) HAPO blend and fuel injection pressure on smoke opacity.

Response Optimizer: Setup

Response	Goal	Lower	Target	Upper	Weight	Importance
NOx (ppmv)	Minimize	98	98	1050	1	1
CO (% vol.)	Minimize	0.07095	0.07095	0.18528	1	1
Smoke (%)	Minimize	10.347	10.347	29.353842	1	1
HC (ppm)	Minimize	16	16	46.424	1	1
EGT (K)	Minimize	458.071	458.071	726.283	1	1
BTHE (%)	Maximize	10.36	28.19599	28.19599	1	1



Figure 17. RSM optimization plot showing predicted output response at selected optimized operating input conditions.

specific optimization criteria and factors. The RSM Response Optimized module was used to design the model and employ desired dimensionless values with a magnitude range from 0 to 1 for assessing the response. If the value of response comes out to be 0, then it is considered unsatisfactory, and scoring the value 1 is considered optimal. The RSM-predicted best optimized input operating conditions were determined to be 40 vol % of HAPO blend ratio, 300 ppm for total binary additive concentration (150 ppm of CuO + 150 ppm of MgO), 441.4 bar for FIP, and 52.32% engine load. The RSM-predicted best optimized output parameters were 20.27% BTHE, 459.357 K for EGT, 16.05 ppm for HC emissions, 11.58% smoke opacity, 0.0927 vol % of CO emissions, and 358.3347 ppm for NO_x emissions within the investigated experimental domain.

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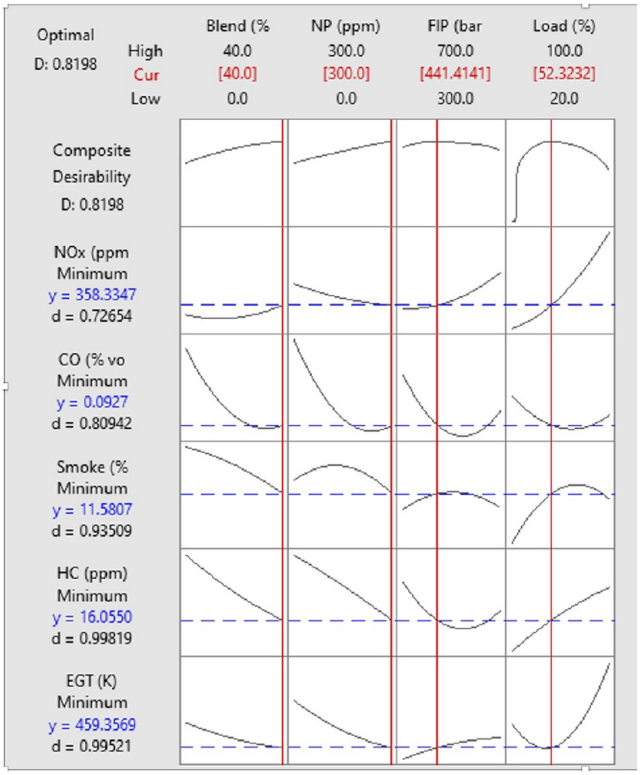


Figure 18. RSM-predicted desirability-based optimization plot with D = 0.8198.

The accuracy of the optimized output parameters was evaluated by performing the experimental verification of the optimization results. The engine input conditions were fixed at these optimized values and a test run was performed for verification. Repeated readings for the same input conditions were observed for the output responses. The percentage error between RSM-predicted values and experimental values for BTHE, EGT, smoke, and CO output responses was less than 5%, whereas for HC and NO_x, the errors were 5.93 and 5.21%, respectively, as shown in the result verification data in Table 12. The signed residuals show that the experimental values were lower than the model prediction for BTHE and higher than the corresponding predictions for EGT, HC, smoke opacity, CO, and NO_x. The differences were of the same order as the model-based SE Fit values, although this comparison is descriptive because condition-specific replicate validation variability was not reported. The CO result is interpreted cautiously because the CO model had limited predicted R^2 performance relatively than other response models. The composite desirability value of the entire optimized experiment was calculated to be 0.8198, which validates the results and can be observed in Figure 18.

5.3. Practical Implications and Limitations of Optimized Operating Conditions

The model determines the 40 vol % HAPO blend and 300 ppm of total binary additive concentration at a fixed 1:1 ratio (150 ppm of CuO + 150 ppm of MgO), fuel injection pressure of 441.4 bar, and 52.32% engine load as optimum only within the short-duration laboratory test conditions and the investigated experimental range. In addition, replicate-level data at the selected optimized conditions were not retained for calculating response-specific experimental standard deviations. The results should not be interpreted as a recommendation for continuous engine operation. A higher concentration of total binary additive at 300 ppm may present practical challenges that were not evaluated in the present study. Sedimentation and dispersion instability factors could cause variation in the concentration of additives in fuel blend delivered to the fuel injection system and may lead to fuel filter clogging, deposits in the injector nozzle, injector wear, ash formation, and may affect other fuel system components over long duration affecting engine durability and storage stability, which were not evaluated. There is also a need for combustion analysis and techno-economic analysis for the feasibility of the experiment under actual conditions.

5.4. Comparison with Recent Studies

The present work findings should be interpreted in the context of recently published investigations on waste-derived oils, fuel additives, and optimization-based engine studies. Direct quantitative comparison among studies should be made cautiously because of varying conditions such as fuel composition, type of additives and their concentration, feedstock, engine configurations, and operating conditions. These factors differ substantially in the investigations.

A recent study in *Industrial& Engineering Chemistry Research*, Yeneneh and Sufe, investigated Al_2O_3 (0−100 ppm) nanoparticles as an additive in waste plastic oil and biodiesel fuel blend in a 1-cylinder diesel engine and performed engine testing with cylinder pressure measurements and ANN modeling.[13] In contrast, the present work assesses a hybrid algae-plastic oil fuel blend with CuO + MgO (0−300 ppm in a 1:1 mass ratio) as a total binary additive. Both studies support the relevance of metal oxide fuel additives for affecting engine performance and emissions. However, the present study cannot directly compare the combustion characteristics with the Al_2O_3-doped WPO study as combustion analysis was not performed.

Table 12. Experimental Verification of Optimized Results: Comparison of RSM-Predicted and Experimental Values of Output Responses

response	optimum RSM-predicted value (P)	experimental value (E)	signed residual ($E - P$)	absolute residual	SE Fit[a]	%error
BTHE (%)	20.27	19.45	−0.82	0.82	1.42	**4.04%**
EGT (K)	459.36	473.8	+14.44	14.44	13.6	**3.14%**
HC (ppmv)	16.05	17	+0.95	0.95	1.95	**5.93%**
smoke opacity (%)	11.58	12.07	+0.49	0.49	0.75	**4.23%**
CO (vol %)	0.093	0.095	+0.0023	0.0023	0.0170	**2.15%**
NO_x (ppmv)	358.34	377	+18.66	18.66	36.0	**5.21%**

[a]SE Fit is the standard error of the RSM model-predicted mean response at the selected operating point obtained from MINITAB. It is not a standard deviation of replicated experimental confirmations.

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For emission reduction comparison, Chetia et al. applied BBD-RSM to a waste cooking oil with CeO_2 and reported a reduction in CO and HC with smoke. Similarly, Uslu et al. examined another metal oxide, Fe_3O_4, with a spirulina microalgae diesel blend and reported an NO_x increment under some conditions. These NO_x variations in the present study support the interpretation that increasing HAPO blend fraction is not only a factor but also interaction effects of blends with additives and combustion phasing contribute as well.

6. CONCLUSION

The detailed study of HAPO-blended diesel fuel with magnesium and copper oxide nanoparticles as additives in a VCR engine was carried out for analyzing their performance and emission characteristics. This study employs a CCD approach in response surface methodology for optimizing the operating input control variables and their interaction effects with output responses. Under the short-duration laboratory conditions, characterized HAPO-blended diesel exhibits stable operation of the test CRDI engine and produces measurable changes in BTHE, EGT, and exhaust emissions. Inclusion of an optimized concentration of 300 ppm combined total binary additive (CuO + MgO) at a fixed mass ratio of 1:1 enhances the BTHE values to 20.27% at optimized input operating parameters, which may indicate improved fuel utility. Addition of nanoparticles has also shown reduced emissions in tested conditions. Practically, the HAPO blend may be suitable for controlled-load stationary diesel engine applications, such as agricultural pump sets and small diesel generators, but long-term durability is to be ensured. This entire experimental study has given the following conclusions:

- The physicochemical properties of the HAPO-blended diesel fuel used were tested as per the ASTM standards, and all properties were within the acceptance limits for use in a diesel engine.
- The combined uncertainty percentage within the experimentation for the different parameters, like temperature sensor, load sensor, etc., was calculated to around ±3.98%.
- CCD-RSM modeling adopted for optimization gave the information about the effect of input variables on output responses. As per the RSM *F*-value, engine load as the input parameter has the greatest effect on the output responses.
- R^2 values from regression analysis were evaluated in the range of 90−99% for BTHE, EGT, smoke, HC, and NO_x output responses, signifying higher accuracy and more reliability of their statistical model. Only the CO model showed limited predictive R^2 performance.
- Surface plots and interaction effects among all of the input variables and output responses were generated using the RSM optimization. The use of CuO and MgO as combined nanoadditives indicates a positive impact on emissions.
- Within the investigated experimental domain, the optimally best input operating conditions were determined to be 40 vol % of HAPO blend ratio, 300 ppm of total binary additive (CuO + MgO) concentration in a 1:1 mass ratio, 441.4 bar for FIP, and 52.32% engine load for which the RSM-predicted best optimized output parameters were 20.27% for BTHE, 459.357 K for EGT, 16.05 ppm for HC emissions, 11.58% smoke opacity, 0.0927 vol % of CO emissions, and 358.3347 ppm for NO_x emissions.
- The overall composite desirability of 82% for the experiment was achieved using the response surface methodology optimization while experimentally verifying the results with an error of less than 5% for BTHE, EGT, smoke, and CO output responses, 5.93% for HC, and 5.21% for NO_x.

These current conclusions are limited by the absence of detailed combustion analysis, including cylinder pressure, heat release rate, ignition delay, combustion duration, and dispersion stability of the nanoparticles. Therefore, the proposed study of performance and emission changes requires verification through combustion analysis and evaluation of fuel consumption through BSFC data.

The entire objective of this experimental study was achieved, which suggests that the hybrid blends of plastic oil and algae oil with the diesel fuel along with CuO and MgO nanoparticles as additives are an optimal combination to be used as an alternative fuel in diesel engines as stable engine operation is achieved without structural modifications to the engine under the experimental conditions employed in this study. With a life cycle assessment and techno-economic analysis, HAPO-blended diesel fuel may provide both economic and environmental pathways for valorization of the plastic waste by reducing landfill burdens and also food-versus-fuel conflicts by using wastewater algae, supporting sustainable biofuel development.

7. FUTURE RESEARCH DIRECTIONS

Further investigations may include the following:

- Other engine control variables were studied, such as the impact of fuel injection timing and EGR on the compression ratio.
- Detailed combustion analysis, fuel consumption, and nanoparticle dispersion stability measurements with varying proportions and nanoparticles' impact on heating value.
- Extended analysis of other specific metal oxide additives and their dispersion stability in the HAPO blends for influence on emission characteristics and heat release rate.
- Various other optimization techniques, like artificial neural networks, fuzzy logics, etc., can be adopted for HAPO blends.
- Tribological studies can be conducted to observe and assess the engine material durability using HAPO blends.
- Techno-economic analysis and life cycle assessment of HAPO fuel blends with nanoparticles could be performed for sustainability evidence.

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Notes

The authors declare no competing financial interest.

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■ ABBREVIATION

ANOVA	analysis of variance
ASTM	American Standard Test Method
BBD	Box−Behnken Design
BTHE	brake thermal efficiency
CCD	central composite design
CO	carbon monoxide
CRDI	common rail direct Injection
EGT	exhaust gas temperature
FIP	fuel injection pressure
HAPO	hybrid algae-plastic oil
HC	unburned hydrocarbons
NO_x	nitrogen oxides
NP	nanoparticle additives
RSM	response surface methodology

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